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The emergence of cognitive digital twin: vision, challenges and opportunities

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ABSTRACT

As a key enabling technology of Industry 4.0, Digital Twin (DT) has been widely applied to various industrial domains covering different lifecycle phases of products and systems. To fully realize the Industry 4.0 vision, it is necessary to integrate multiple relevant DTs of a system according to a specific mission. This requires integrating all available data, information and knowledge related to the system across its entire lifecycle. It is a challenging task due to the high complexity of modern industrial systems. Semantic technologies such as ontology and knowledge graphs provide potential solutions by empowering DTs with augmented cognitive capabilities. The Cognitive Digital Twin (CDT) concept has been recently proposed which reveals a promising evolution of the current DT concept towards a more intelligent, comprehensive, and full lifecycle representation of complex systems. This paper reviews existing studies relevant to the CDT concept, and further explores its definitions and key features. To facilitate CDT development, a reference architecture is proposed based on the RAMI4.0 and some other existing architectures. Moreover, some key enabling technologies and several application scenarios of CDT are introduced. The challenges and opportunities are discussed in the end to boost future studies.

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1. Introduction

Since its first appearance in 2003 (Grieves 2014), Digital Twin (DT) has been a popular topic for both academia and industry. It is recognised as a key enabling technology for realising the paradigm of smart manufacturing and Industry 4.0 (Tao, Zhang et al. 2019). The DT vision refers to a comprehensive physical and virtual description that captures all relevant attributes and behaviours of a component, product or system (Boschert, Heinrich, and Rosen 2018; Abburu et al. 2020b). To avoid misunderstanding, in this paper we use the term ‘system’ to represent the targeted physical component, product or system of a DT, considering that most modern industrial components and products can be treated as a system. A basic DT model is composed by at least three elements including the physical entities in real world, virtual entities in virtual space, and the interactions between physical and virtual entities. Some recent studies (Tao et al. 2017; Qi et al. 2019) consider DT data and services as separate elements to create an extended version of the basic three-dimension DT model. Empowered by the rapid advancement of information technologies and the extensive deployment of Industrial Internet of Things (IIoT), DTs have been applied to almost all industrial sectors covering different lifecycle phases of a system such as

concept, design, production, and maintenance, etc. Tao, Zhang et al. (2019).

Modern industrial systems are getting increasingly complex due to the continuously adoption of advanced technologies. DTs are considered as key digital assets across the entire lifecycle of a system, thus, it is necessary to comprehensively network and integrate all relevant DT models of the system (Tao, Zhang et al. 2019). In addition, a complex system usually contains multiple subsystems and components which may have their own DT models. These DT models might be created by various stakeholders based on different protocols and standards whose data structures are usually heterogeneous in terms of syntax, schema, and semantics. This makes the integration of DT models a challenging task.

Semantic technologies have been used as key enabling components in many intelligent systems to achieve semantic interoperability for heterogeneous data and information (Cho, May, and Kiritsis 2019; Psarommatidis 2021). Semantic models enable to capture system information in an intuitive way and to provide a concise and unified description of such information. They describe the information in standardised ontology languages making it possible to specify direct interrelationships among various systems and models (Kharlamov

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et al. 2018). As an advanced semantic technology, knowledge graph enables to describe model information in the form of entities and relationships, which makes it possible of creating new knowledge using a reasoner (Ehrlinger and Wöß 2016; Nickel et al. 2015). This makes semantic modelling and knowledge graph modelling promising solutions for integrating heterogeneous DT models involved in a complex system across different domains and lifecycle phases.

During the past few years, there has been a trend to combine DT with advanced semantic modelling technologies to provide DT with cognition capabilities. A new concept, named Cognitive Digital Twin (CDT), or Cognitive Twin (CT) in some cases, has appeared in several recent studies, which is considered as a promising evolution trend of DT. We use the acronym CDT to represent this concept in the following sections to avoid confusion. The aim of this paper is to review-related works and explore the CDT vision and features. Moreover, we propose a reference architecture and introduce some application scenarios and key enabling technologies for realising the CDT vision. The challenges and opportunities are also discussed to pave way for future studies.

2. Related works

2.1. Digital twin and its application in production

Owing to its unique advantages in connecting the physical and virtual spaces of industrial systems, DT has attracted enormous attention from both industry and academic since its appearance. During the early stage of DT development, it was primarily applied to support prognostics and health management (PHM) (Tao, Sui et al. 2019), such as aircraft structural health management and flight state assessment (Gockel et al. 2012; Seshadri and Krishnamurthy 2017). With the rapid development of Industry 4.0 and relevant technologies, DTs have been widely applied in different domains, among which production is one of the main focuses. These applications correspond to different lifecycle phases of a product or a production system, such as product/system design phase, production and operation phase, and recovery or remanufacturing phase, etc.

The production and operation phase covers the most typical DT applications including the aforementioned PHM, predictive maintenance, production system optimisation etc. For example, Tao et al. (2017) proposed a digital twin shop-floor concept and discussed its characteristics, architecture and operating mechanism etc. Söderberg et al. (2017) utilised DT to assure real-time product geometry during individualised production.

Knapp et al. (2017) investigated the key building blocks of DT for the additive manufacturing processes and verified its capabilities for predicting properties and serviceability of components. Wang, Ye et al. (2019) proposed a DT reference model to support fault diagnosis of rotating machinery. Ding et al. (2019) combined the Cyber-Physical System (CPS) and DT technologies to create a DT-based Cyber-Physical Production System in smart shop floors aiming at autonomous manufacturing.

Many DT applications can also be found focusing on the design and development phase of products or production systems. For example, Schleich et al. (2017) proposed a reference model for the design phase of products based on the concept of Skin Model Shapes to facilitate model conceptualisation, representation, and implementation. Tao, Sui et al. (2019) proposed a DT-driven framework which applied DT to support product design. There are relatively less DT applications focusing on the end of the lifecycle. Wang and Wang (2019) proposed a DT-based system for recycling, recovery and remanufacturing of waste electrical and electronic equipment, which makes possible of full lifecycle support for manufacturing/remanufacturing operations from design to recovery.

The review of related works indicates that most existing DT applications focus on a single lifecycle phase of the corresponding system. However, to realise the vision of Industry 4.0 and smart manufacturing, it is necessary to integrate DTs of different lifecycle phases of the same system. It is a challenging task due to the high complexity of modern industrial systems. To cope with this challenge, some recent studies converge semantic modelling, systems engineering with DTs to enhance their interoperability and connectivity. It makes possible of certain cognitive and autonomous capabilities for DTs.

2.2. The emergence of cognitive digital twin

Previous studies have explored the feasibility of enhancing DTs' cognitive capabilities using semantic technologies. Kharlamov et al. (2018) proposed a concept of semantically enhanced DTs based on semantic modelling and ontologies. It enables to capture the characteristics and status of a system, as well as how it interacts with other components in a complex system. Gómez-Berbís and de Amescua-Seco (2019) created a Semantic Digital Twin (SEDIT) based on IoT data management and knowledge graphs across the entire enterprise to provide formal representations of the domain specific DTs. It enables to find data spread across complex systems, and to provide inferences dynamically in ad-hoc and task-oriented frameworks. Banerjee et al. (2017) developed a

query language based on knowledge graphs for extracting and inferring knowledge from large scale production data. It verifies the feasibility of knowledge graph technology to support DT development and management in complex systems. Boschert, Heinrich, and Rosen (2018) investigated a paradigm of the next generation DTs, for which knowledge graph was considered as a main enabling technology for linking and retrieving heterogeneous data, as well as descriptive and simulation models.

These aforementioned studies have attempted to enhance the cognition capabilities of DTs with semantic technologies, although not formally using the term *cognition* or *cognitive*. The term '*Cognitive Digital Twins*' firstly appeared in the industry sector proposed by Adl (2016). During an industry workshop presentation in 2016, Ahmed El Adl discussed about the cognitive evolution of IoT technologies and proposed the Cognitive Digital Twin concept as well as its characteristics and categories. It was defined as '*a digital representation, augmentation and intelligent companion of its physical twin as a whole, including its subsystems across all of its life cycles and evolution phases*'. Fariz Saracevic (2017) from IBM also presented a similar concept later during a workshop in 2017. He approached this concept from the perspectives of cognitive computing and artificial intelligence, and used IBM Watson as examples to demonstrate the cognitive engineering scenarios. Compared with the previous definition, some specific expected functions are proposed in this new definition, i.e. '*CDT uses real time data from IoT sensors and other sources to enable learning, reasoning and automatically adjusting for improved decision making*'.

In addition to the efforts from industry sector, researchers from academia are also contributing to promote the CDT concept. Boschert, Heinrich, and Rosen (2018) prospected the paradigm of next generation digital twin (nexDT). They claim that the current isolated DT models cannot fulfil all purposes and tasks across the entire lifecycle and the integration of multiple DT models for different business objectives is needed. Semantic technologies like knowledge graph are promising tools to connect Product Lifecycle Management (PLM) systems, cloud solutions and further data artefacts as well as devices. The main elements of nexDT definition align well with the CDT concept. For example, nexDT contains a set of semantically linked digital artifacts, including design and engineering data, operational data and behavioural descriptions; the nexDT is expected to contain all available and required data and knowledge evolved with the real system in different lifecycle phases.

Fernández et al. (2019) explores the functions of hybrid human-machine cognitive systems and specified

the Symbiotic Autonomous Systems (SAS). The authors considered CDT as '*a digital expert or copilot, which can learn and evolve, and that integrates different sources of information for the considered purpose*'. SAS focuses on the cooperation, and convergence of human and machine augmentation, with increasing intelligence and consciousness, leading towards a continuous symbiosis of humans and machines. Based on the SAS context, the authors created a Associative Cognitive Digital Twin (AC-DT) framework to enhance the applications of CDTs. AC-DT is a contextual augmented description of an entity which aims at a specific cognitive purpose and includes the relevant associated connections with other entities.

Lu, Zheng et al. (2020) proposed a formal definition of Cognitive Twins as '*DTs with augmented semantic capabilities for identifying the dynamics of virtual model evolution, promoting the understanding of inter-relationships between virtual models and enhancing the decision-making*'. They developed a framework based on knowledge graph to support the development of the CDTs. To facilitate the application of CDTs, they provided a tool-chain consisting of multiple existing software and platforms to empower different components of the CDT models. They verified the proposed CDTs framework and related tool-chain through an application case for supporting decision-makings of an IoT system development.

Abburu et al. (2020b) reviewed existing concepts related to digital twins and provided a three-layer approach to clarify different 'twins' into digital twins, hybrid twins and cognitive twins. The definitions are according to the capabilities of the 'twins': Digital twins are isolated models of physical systems; Hybrid twins are interconnected models capable of integrative prediction of unusual behaviours; Cognitive twins are extended with expert and problem-solving knowledge capable of dealing with unknown situations. Based on these definitions, a CDT should incorporate cognitive features to enable sensing complex and unpredicted behaviours; and reasoning for optimisation strategies leading to a system that continuously evolving. A five-layer implementation architecture is also proposed in this study containing a software toolbox, which is applied to several relevant use cases in the process industry to verify its possible applicability.

Aiming at creating smart manufacturing systems, Ali et al. (2021) proposed to extend existing DTs with additional intelligent capabilities supported by a three-layer architecture i.e. access, analytic and cognition. The cognitive layer is enabled by edge computing, domain expertise, and global knowledge bases. It supports the integration of multiple DTs by building customised

communication networks thus to perform autonomous decision-makings. It also envisioned an ecosystem of CDTs by connecting a large number of CDTs of different systems and domains.

In a recent study, Al Faruque et al. (2021) proposed the CDT for manufacturing systems based on the advances of cognitive science, artificial intelligence technologies and machine learning techniques. According to the fundamental aspects of cognition, they define CDT as digital twin with additional cognitive capabilities including: perception (*forming useful representations of data related to the physical twin and its physical environment*), attention (*focusing selectively on a task or a goal or certain sensory information either by intent or driven by environmental signals and circumstances*), memory (*encoding information, storing and maintaining information, and retrieving information*), reasoning (*drawing conclusions consistent with a starting point*), problem-solving (*finding a solution for a given problem or achieving a given goal*), and learning (*transforming experience of the physical twin into reusable knowledge for a new experience*). The potential applications of CDT in the product design phase include: searching and discovering digital models and twins of products, processes and systems; sharing information and knowledge among digital twins; and scale knowledge transferring across multiple domains.

The review of related work indicates that the CDT is still in its infancy. Existing studies have been attempting to extend existing DT models from different perspectives using various advanced technologies. Some recent works have provided formal definitions and architectures for CDT implementation. In the following sections, we further explore and summarise the CDT paradigm from perspectives of its vision, reference architecture, applications, enabling technologies, as well as challenges and opportunities.

3. Cognitive digital twin vision

3.1. Cognitive digital twin characteristics and definition

The CDT concept has been formally defined in several existing works. These existing definitions are listed in Table 1 for comparison in order to derive a comprehensive definition. Although there is no widespread consensus on the CDT definition, some common elements and features can be extracted.

- (1) DT-based: CDT is an extended or augmented version of DT. It contains at least the three basic elements of DT, including the physical entity (systems, subsystems, components etc.), digital (or virtual) representation or shadows, the connections between the virtual and physical spaces. The main difference is that CDT usually contains multiple DT models with unified semantics topology definitions. Particularly for a complex industrial system, its CDT should include digital models of its subsystems and components, and each of them has different status across the entire lifecycle. A large number of related digital models are expected to be connected for more complex business scenarios.
- (2) Cognition capability: As indicated in its name, a CDT should have certain cognition capabilities, meaning it enables to perform human-like intelligent activities such as attention, perception, comprehension, memory, reasoning, prediction, decision-making, problem-solving, reaction and so on Kokar and Endsley (2012), Fernández et al. (2019) and Al Faruque et al. (2021). Thus, a CDT is defined to recognise complex and unpredicted behaviours with optimisation strategies dynamically. Although it is still far from fully realising this target, the fast development of semantic technologies, artificial intelligence, IIoT and ubiquitous sensing technologies etc. make it possible to realise cognition capabilities at a certain level.
- (3) Full lifecycle management: A CDT should consist of digital models covering different phases across the entire lifecycle of the system, including beginning-of-life (BOL, e.g. design, building, testing), middle-of-life (MOL, e.g. operating, usage, maintenance) and end-of-life (EOL, e.g. disassembly, recycling, remanufacturing). It should also be capable of integrating and analysing all available data, information and knowledge from different lifecycle phases thus to support aforementioned cognitive activities.
- (4) Autonomy capability: A CDT should conduct autonomous activities without human assistance or minimum level of human intervention. This capability is partially overlapped with and empowered by the cognition capabilities of a CDT. For example, based on the perception and prediction results, a CDT can autonomously make decisions and react for design, production or operations adaptively.
- (5) Continuous evolving: A CDT should be able to evolve with the real system along the entire lifecycle. There exists three levels of evolving. First, for a single digital model, it updates itself according to the change of relevant data, information and knowledge from the real system; second, due to the interactions among different digital models contained in the same lifecycle phase, each model evolves dynamically according to the impact of other models; third, due to the feedback from other lifecycle phases, the

Table 1. Definitions of Cognitive Digital Twin and similar concepts from related works.

Authors	Concept	Definition
Adl (2016)	Cognitive Digital Twin	'A digital representation, augmentation and intelligent companion of its physical twin as a whole, including its subsystems and across all of its life cycles and evolution phases.'
Fariz Saracevic (2017)	Cognitive Digital Twin	'A virtual representation of a physical object or system across its lifecycle (design, build, operate) using real time data from IoT sensors and other sources to enable learning, reasoning and automatically adjusting for improved decision making'
Boschert, Heinrich, and Rosen (2018)	Next Generation Digital Twin	'A description of a component, product, system or process by a set of well-aligned, descriptive and executable models' which 'is the semantically linked collection of the relevant digital artefacts including design and engineering data, operational data and behavioural descriptions' and 'evolves with the real system along the whole life cycle and integrates the currently available and commonly required data and knowledge'
Matthews (2018)	Cognitive Digital Twin	'A digital representation and augmentation of an entity across its lifecycle to optimise cognitive activities'
Fernández et al. (2019)	Associative Cognitive Digital Twin (AC-DT)	'A contextual augmented description that explicitly includes the relevant associated relationships, connections or information channels of the considered entity with the other entities, for the considered application and purpose'
Eirinakis et al. (2020)	Enhanced Cognitive Twin (ECT)	'Introduce advanced cognitive capabilities to the DT artefact that enable supporting decisions, with the end goal to enable DTs to react to inner or outer stimuli. The ECT can be deployed at different hierarchical levels of the production process, i.e. at sensor-, machine-, process-, employee- or even factory-level, aggregated to allow both horizontal and vertical interplay'
Lu, Zheng et al. (2020)	Cognitive Twin	'Digital Twins with augmented semantic capabilities for identifying the dynamics of virtual model evolution, promoting the understanding of interrelationships between virtual models and enhancing the decision-making based on DT'
Abburu et al. (2020b)	Cognitive Digital Twin (COGNITWIN)	'An extension of Hybrid Digital Twins incorporating cognitive features that enable sensing complex and unpredicted behaviour and reason about dynamic strategies for process optimisation, leading to a system that continuously evolve its own digital structure as well as its behaviour'
Ali et al. (2021)	Cognitive Digital Twin	'an extension of existing digital twins with additional capabilities of communication, analytics, and intelligence in three layers: i) access, ii) analytics, and iii) cognition.'
Al Faruque et al. (2021)	Cognitive Digital Twin	'Digital Twins endowed with the other elements of cognition such as perception, attention, memory, reasoning, problem-solving, etc.'

previous two situations may happen simultaneously, even new models and components will be added.

On the basis of existing CDT definitions and the key features listed above, this paper attempts to define CDT as:

Cognitive Digital Twin (CDT) is a digital representation of a physical system that is augmented with certain cognitive capabilities and support to execute autonomous activities; comprises a set of semantically interlinked digital models related to different lifecycle phases of the physical system including its subsystems and components; and evolves continuously with the physical system across the entire lifecycle.

It is worth to clarify that the above definition focuses on the virtual space of the 'twins' to align with existing studies. In the rest of the paper, we use the singular form CDT to represent a complete twin system including both physical and virtual spaces. The reason is to avoid misunderstanding and follow the commons of digital twin studies. Whereas the plural form CDTs is used to represent multiple independent cognitive digital twin systems or implementation cases.

3.2. Digital twin versus cognitive digital twin

According to the previously mentioned CDT definitions, it is obvious that there are both common features and

differences between a traditional DT and CDT. It is necessary to clarify their similarities and differences in order to facilitate the development of both concepts. A general comparison of both concepts is depicted in Figure 1.

The CDT concept is built on the basis of DT. Therefore, a CDT should include all the essential characteristics of DT. For example, both DT and CDT enable digital representations of a physical system, meaning that they both contain the three basic elements: (1) physical entities in the real space; (2) virtual entities in the virtual space; (3) the connections between physical and virtual entities. Moreover, the virtual entities of both DT and CDT can communicate with their corresponding physical systems and update dynamically. From this point of view, the CDT concept can be treated as a subset of DT, as depicted in Figure 2. It means that all CDTs are certain kind of DT with extended characteristics such as cognitive capabilities, cross-lifecycle phases and multiple system levels.

The main differences between them are their structure complexity and cognitive capability. First, a CDT is usually more complex than a DT in terms of architecture and number of lifecycle phases involved. Most DTs correspond to a single system (or product, subsystem, component etc.), and focus on one of the lifecycle phases. In contrast, a CDT should consist of multiple digital models corresponding to different subsystems and

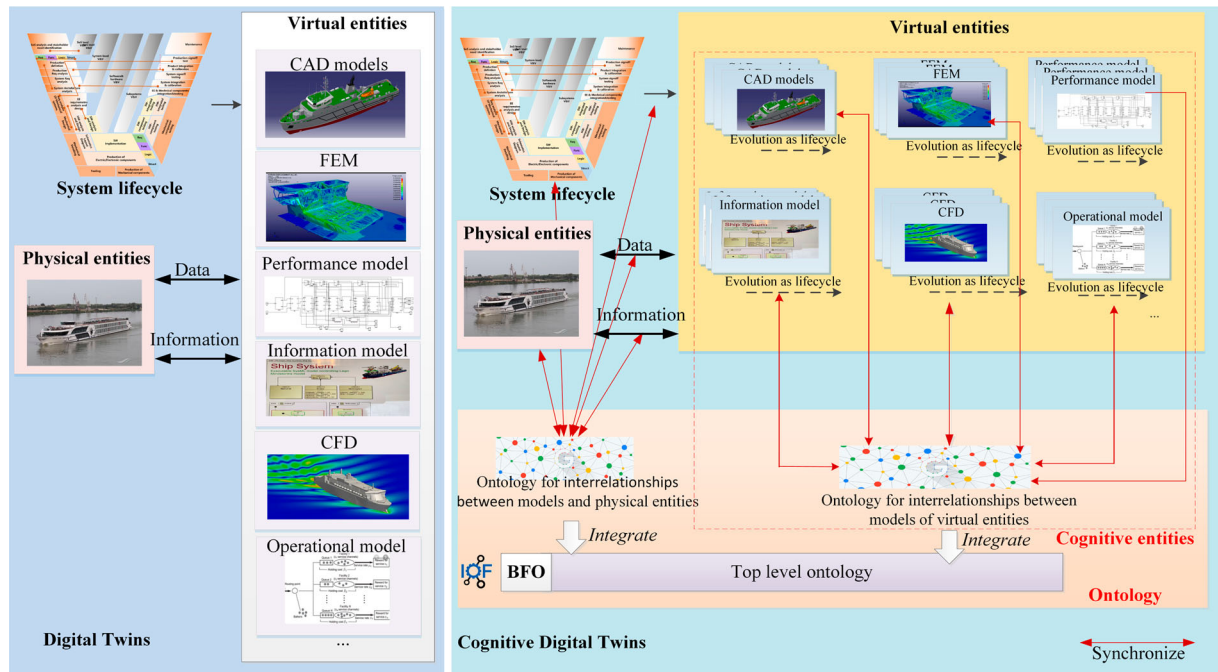


Figure 1. Comparison between digital twins and cognitive digital twins.

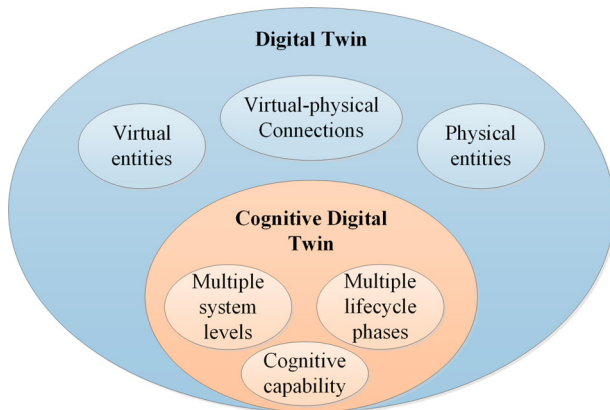


Figure 2. Relation between digital twin and cognitive digital twin.

components of a complex system, and focus on multiple lifecycle phases of the system. In many cases, a CDT can be constructed by integrating multiple related DTs using ontology definition, semantic modelling and lifecycle management technologies. These DTs may correspond to different subsystems and components mapped to different lifecycle phases, and each of them evolves along with the system lifecycle.

Second, cognitive capabilities are essential for CDTs, whereas DTs do not necessarily possess such capabilities. Most existing DTs are applied for services of visibility, analytic and predictability, such as condition monitoring, functional simulation, dynamic scheduling, abnormal detection, predictive maintenance and so on. These

services are usually enabled by data-based and model-based algorithms using the data collected from the physical entities. Cognitive capability is required to reach higher level automation and intelligence, for example, to enable sensing complex and unpredicted behaviours and generating dynamic strategies autonomously. To achieve this target, the data-based and model-based algorithms are incapable of integrating the complex data and models from different systems and lifecycle phases with heterogeneous specifications and standards. To address this challenge, it requires more technologies such as semantic modelling, systems engineering and product lifecycle management and so on. For instance, a unified ontology can represent physical entities, virtual entities and the topology between them which is the basis to realise the cognitive capability. Top level ontologies can be used to integrate different ontologies synchronised with virtual entities across the lifecycle in order to support reasoning for cognitive decision-makings.

It is important to emphasise that, the CDT concept is not proposed to replace DT, instead, it is an extended and federated version of the current DT paradigm. A trade-off should be made according to the requirements of the application scenarios and stakeholders' requirements. CDTs aim at complex systems that consist of multiple interlinked subsystems and components, and require interactions between different lifecycle phases. It can provide more advanced cognitive capabilities, whereas the implementation is more challenging in terms of technology readiness, risk level, economic and time cost

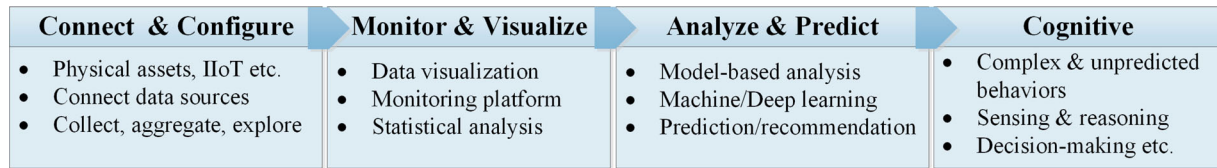


Figure 3. Cognitive engineering journey (adapted from Fariz Saracevic 2017).

etc. In comparison, the enabling technologies of DT are more mature, and many successful cases are available as references.

The cognitive engineering journey, as shown in Figure 3 (Fariz Saracevic 2017), provides a functional perspective to help determine when a digital twin should be augmented with cognitive abilities. A typical digital twin covers the first three functions, i.e. connect and configure of the physical system, monitor and visualisation based on collected data, analyse and predict using model-based or data-driven approaches. These functions enable typical DT services such as process monitoring, abnormal detection, predictive maintenance etc. Beyond these functions, cognitive capabilities will be necessary, for example when it's required to deal with unpredicted behaviours of dynamic systems, or to perform sensing and reasoning for autonomous decision-making etc. Another perspective is from the complexity of the physical system. A digital twin is capable of representing a system, for example a machine, an assembly line or an entire factory in manufacturing domain. CDTs target at more complex scenarios when multiple systems are involved, especially when they have different data standards and specifications. In many cases, they also correspond to multiple lifecycle phases of the system.

3.3. Cognitive digital twin reference architecture

A reference architecture aims to capture the essence of a collection of system architecture in the same domain, thus to guide the development of new architectures of systems from this domain (Muller 2008). In order to facilitate the development of CDT, a couple of reference architectures for CDT have been proposed in previous studies.

3.3.1. Reference architecture overview

A reference architecture framework was proposed by Adl (2016) which specifies the key architectural building blocks of CDT in its early development phase. According to this framework, the digital representations of the physical entities in the virtual space are based on the Cognitive Digital Twin Core (CDTC), which contains metadata, self-defense mechanisms and governing rules to empower the CDT functions. As shown in Figure 4, the

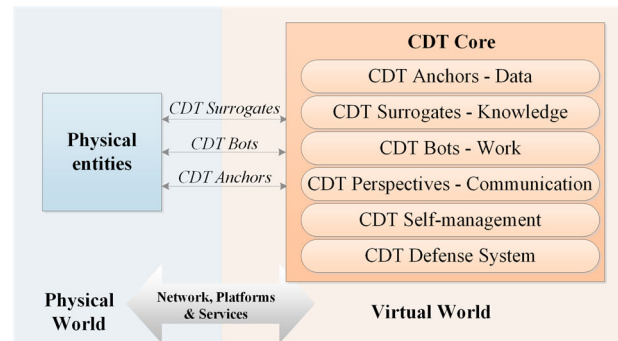


Figure 4. CDT reference architecture framework (adapted from Adl 2016).

CDTC is composed of six layers including anchors (data workers), surrogates (knowledge workers), bots (makers), perspectives (interfaces), self-management (administrators), and defense systems (guardians). This architecture framework summarises the key functions that a CDT should support and the enabling mechanisms.

Lu, Zheng et al. (2020) designed a knowledge-graph-centric framework for CDT according to the standard ISO/IEC/IEEE 42010 (Duprez 2019). It consists of five main components covering multiple domains including industrial system dynamic process modelling, ontology-based cross-domain knowledge graphs, CDT construction for dynamic process simulation, CDT-based analysis for process optimisation and service-oriented interface for data interoperability. This framework is designed for supporting decision-makings during IoT system development. It takes inputs from business domains and provides outputs to asset domains. Abburu et al. (2020b) proposed an architectural blueprint for CDT that consists of five layers providing a set of model-driven and data-driven services. The five layers include data ingestion and preparation layer, model management layer, service management layer, twin management layer and user interaction layer. Moreover, to better standardise the proposed architecture, its layers and components are mapped to the reference models of Big Data Value Association (BDVA) reference model and the Artificial Intelligence Public Private Partnership (AI PPP). Several use cases are also introduced to demonstrate the ability and versatility of the proposed architecture.

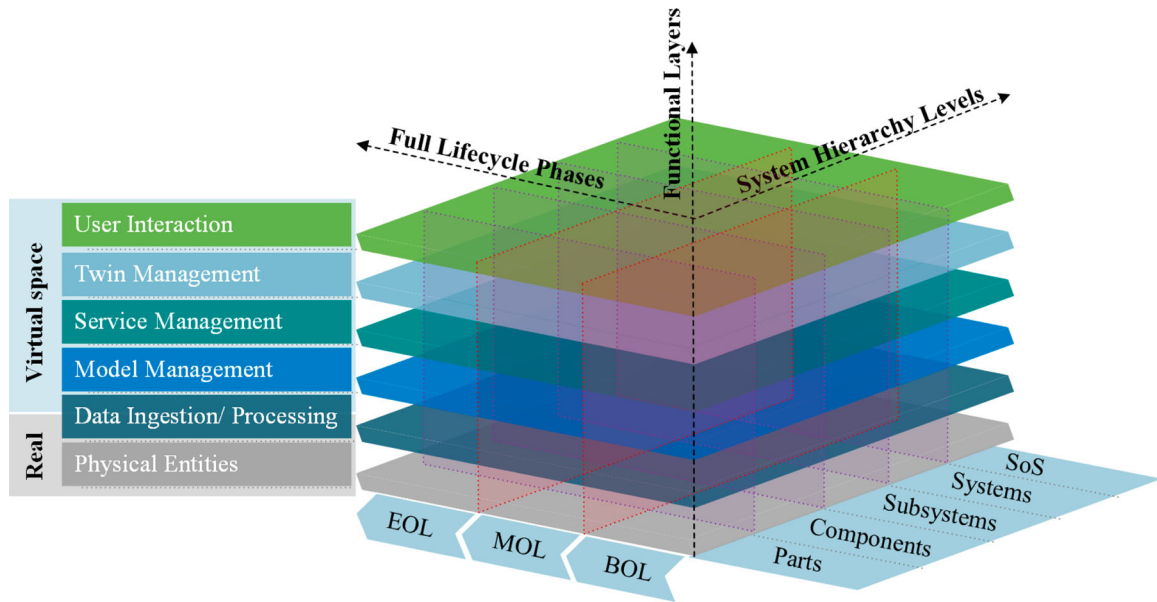


Figure 5. CDT reference architecture based on RAMI4.0.

These existing architectures fit well with the corresponding definitions of CDT and cover different elements and characteristics of the summarised definition aforementioned. Based on these existing architectures, we attempt to design a more comprehensive reference architecture for CDT aiming at covering all its key elements and characteristics. One of the principles for developing a new reference architecture is to reuse and comply existing standards and specifications to assure the interoperability. RAMI4.0 (Schweichhart 2016) is a widely adopted reference architecture for industry 4.0 digitisation. It provides a three-dimensional framework covering the most important aspects of Industry 4.0, including the lifecycle value stream, six layers of common business scenarios and the communication layers corresponding to a smart factory hierarchy. This three dimensions overlaps some of the key elements of the CDT definition, especially the lifecycle value stream dimension and the multi-layer approach for structuring physical systems and communication levels.

Inspired by the RAMI4.0 reference architecture and taking into account the existing CDT architectures, we designed a three-dimensional CDT reference architecture covering the key elements of the CDT definition. As shown in Figure 5, the three dimensions of the proposed reference architecture include full lifecycle phases, system hierarchy levels and six functional layers. The elements of each dimension are explained below:

- **Full lifecycle phases:** This dimension focuses on the fully lifecycle management and continuous evolving

capabilities of CDT definition. During the entire lifecycle of a system, many digital models are created to support different lifecycle phases along this dimension. For example, a typical lifecycle of a manufactured product includes production design, simulation, manufacturing process planning, production, maintenance and recycling etc. Each of these phases may have multiple related digital models. It is worth noting that a CDT does not necessarily contain digital models covering all lifecycle phases, but it is supposed to support the integration of models across different lifecycle phases.

- **System hierarchy levels:** This dimension provides a hierarchical approach to specify the structure and boundary of a CDT. Modern industrial systems are usually highly complex system-of-systems (SoS). It is crucial to properly define the scope of a CDT. When developing CDTs, it might be difficult to create the models in a separated way simply according to the physical hierarchy of a factory as specified in RAMI4.0. Therefore, we adopted a systems engineering methodology to define the structure of a complex into SoS, system, subsystem, components and parts (Kossiakov et al. 2011).
- **Functional layers:** This dimension specifies the different functions provided by a CDT. It is designed based on the architectural blueprint of the COGNITWIN Toolbox (CTT) proposed by Abburu et al. (2020b). The Physical Entities layer represents the system in the physical space, and the other five layers represent different functions of CDT in the digital space, including

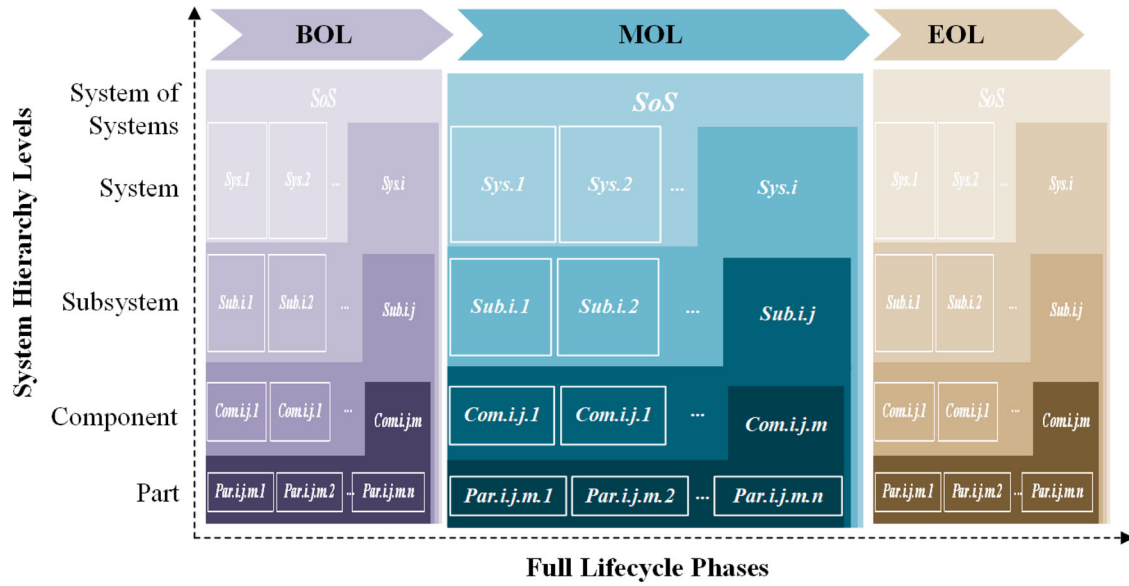


Figure 6. The full lifecycle phases and system hierarchy levels plane of CDT reference architecture.

Data Ingestion and Processing, Model Management, Service Management, Twin Management and User Interaction.

3.3.2. Reference architecture components

In order to explain the components of the proposed reference architecture, the three-dimensional architecture is decomposed into two-dimensional planes.

- (1) The perspective of Lifecycle phases and System hierarchy levels: This perspective shows how the digital models are identified corresponding to system hierarchy levels and lifecycle phases. As shown in Figure 6, a complex system is treated as a *System of systems* (SoS) which consists of several systems (Sys.1, Sys.2, ..., Sys.i). Each system contains multiple subsystems, e.g. Sys.i contains (Sub.i.1, Sub.i.2, ..., Sub.i.j). Each subsystems can be further decomposed into a set of components, e.g. Sub.i.j to (Com.i.j.1, Com.i.j.2, ..., Com.i.j.m); and each component can be finally decomposed into multiple parts, e.g. (Par.i.j.m.1, Par.i.j.m.2, ..., Par.i.j.m.n). Each of these elements corresponds to one or more digital models which are dynamic and evolving. As shown in Figure 6, there are different versions of the models during each lifecycle phase. It is worth to notice that, although the models of different phases are discrete in Figure 6, in reality these models are all linked and related which means the evolving of models leads to the changes of other model versions.
- (2) The perspective of Functional layers and System hierarchy levels: This perspective shows the main function blocks of a CDT categorised as different

functional layers as shown in Figure 7. This structure is organised based on previous studies (Abburu et al. 2020b; Lu, Zheng et al. 2020). The *Physical Entities* layer represents the real systems in the physical space. The data sources located at this layer generate different types of data and information reflecting the status of different hierarchy levels of the physical system. These data are then transferred to the data and meta-data repositories at the *Data Ingestion & Processing* layer based on certain protocols. This data ingestion process are realised via specific brokers and adapters. Based on the requirements of the corresponding services, these data can be processed and analysed using data mining techniques supported by edge/fog/cloud computing.

The *Model Management Layer* uses semantic technologies, such as knowledge graphs and ontology models, to store and manage different types of DT models, such as first-principle models, empirical models and knowledge-driven models. A model manager is included in this layer to manage the models in model repository. It supports: (1) version management, referring to track and control changes to a collection of related models; (2) traceability management, referring to the planning, organisation, and coordination of all the models and their operations related to traceability; (3) consistency management, referring to support in consistency violation identification and resolution to ensure that the models are syntactically and semantically correct; (4) co-design, the act of creating with stakeholders within the entire lifecycle to ensure the developed models meet their needs; (5) process management, the

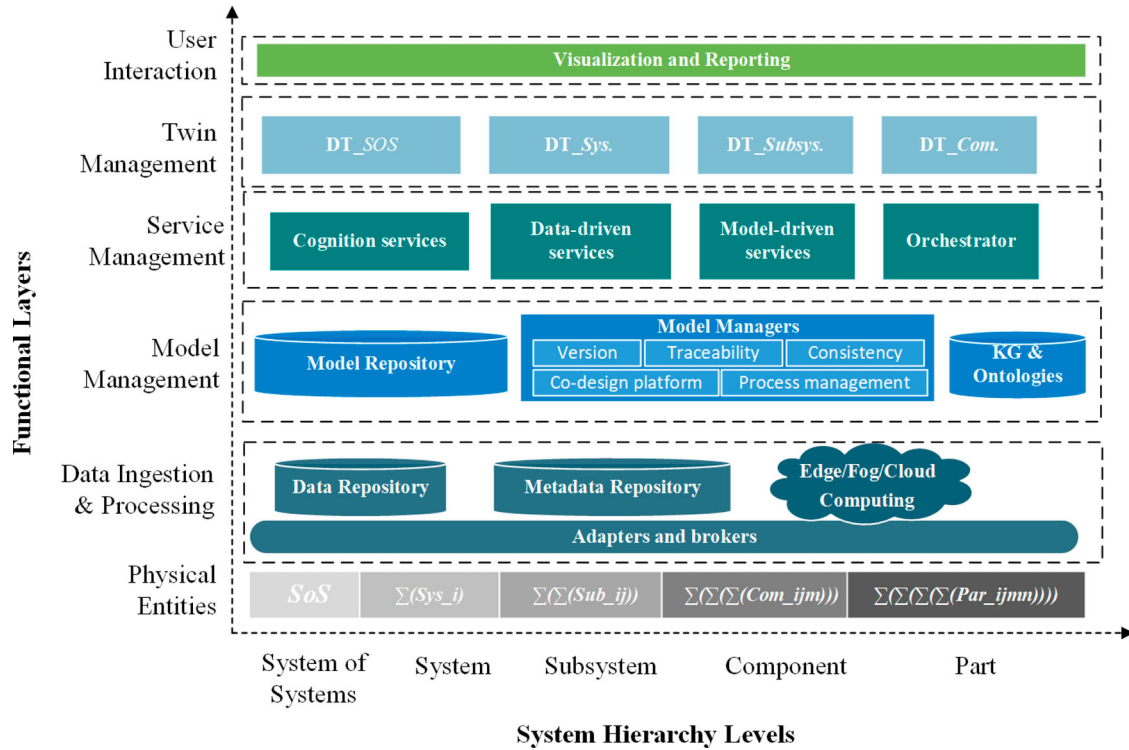


Figure 7. The functional layers and system hierarchy levels plane of CDT reference architecture.

perspective in which stakeholder use various methods to discover, model, analyse, measure, improve, optimise, and automate modeling processes.

According to the requirements of the application scenarios, different business services such as data-driven services, model-driven services and cognition services etc., are registered and orchestrated at the *Service Management* layer. Multiple DTs can be involved in a CDT corresponding to different systems, subsystems and components. Each of these DTs evolves along with time. The *Twin Management* layer aims to support the updating and orchestration of these DTs across the entire lifecycle. On top of all the functional layers, it is the *User Interaction* layer to enable the stakeholders of the CDT to interact with the CDT through front-end. The interrelationships between function blocks and layers are not depicted in the figure since this paper does not aim to analyse the details of each function. More detailed explanation can be found in the aforementioned papers (Abburu et al. 2020b; Lu, Zheng et al. 2020).

- (3) The perspective of Lifecycle phases and Functional layers: This perspective shows how the functions at each layer are executed across different lifecycle phases. As shown in Figure 8, raw data are generated from different lifecycle phases of a system. The lifecycle phases are categorised into BOL, MOL and EOL, which includes more detailed phases ($t1$

in BOL, $t2$ in MOL and $t3$ in EOL). For example, $t1$ refers to design and testing; $t2$ includes operating and maintenance; and $t3$ consists of disassembly and re-manufacturing etc. The data from different phases are transferred into the data repository and meta data repository, and then to be processed with proper data mining techniques such as machine learning. Based on the processed data, corresponding models are developed and stored in the model repository. Moreover, a model manager enables to support related management activities for the entire lifecycle of related models. At the Service management layer, data-driven and model-driven services are provided focusing on one single lifecycle phase or cross-phase services. At the Twin management layer, the digital twins corresponding to different system hierarchy levels are updated according to the new data and knowledge.

This reference architecture aims to provide a guidance for designing a general conceptual structure of a CDT. It requires many advanced technologies to develop a functional CDT and apply it to real industrial scenarios.

3.3.3. Compare with relevant reference architectures

3.3.3.1. Compare with RAMI4.0. The proposed CDT reference architecture adopted RAMI4.0 as the foundation

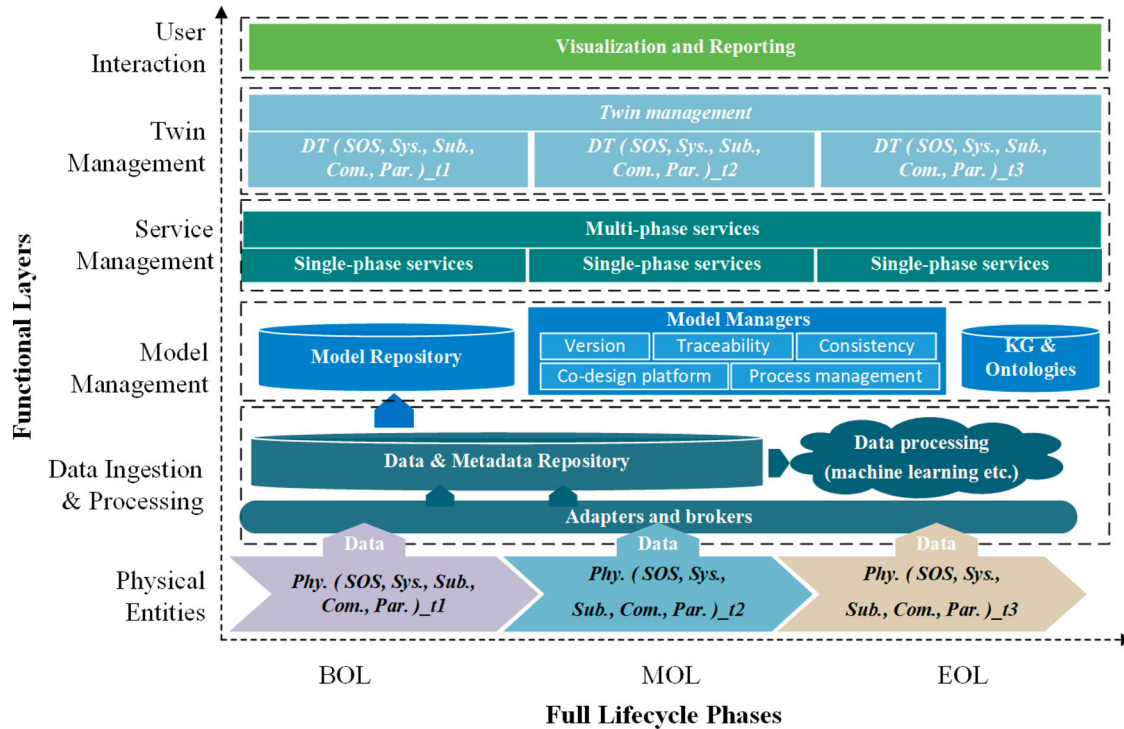


Figure 8. The functional layers and lifecycle phases plane of CDT reference architecture.

to construct the three main pillars. The *Lifecycle Phases* of the CDT remains equivalent to the *Life Cycle Value Stream* (IEC 62890) dimension of the RAMI4.0. The RAMI4.0 *Hierarchy Levels* (IEC 62264//IEC 61512) dimension is replaced by the *System Hierarchy Levels* in CDT, which decomposed the system hierarchy from systems engineering perspective. The RAMI4.0 *Hierarchy Levels* mainly focus on the physical system hierarchy, including product, field device, control device, station, work centres, enterprise and connected world. In contrast, the CDT *System Hierarchy Levels* covers both physical systems and processes, which allows the application of systems engineering technologies such as MBSE to facilitate CDT development.

The *Architecture Layers* of the RAMI4.0 is replaced by the *Functional Layers* in the CDT architecture as shown in Figure 9. Although they are both composed of physical and digital spaces, their virtual entities are different. The RAMI4.0 *Architecture Layers* decomposes the layers using a business-oriented strategy. It aims to answer the basic questions about realising a business idea by specifying organisation and business processes, functions of the asset, necessary data, access to information, and integration of assets into the real world through the digitalisation layer (Schweichhart 2016). In contrast, the CDT *Functional Layers* focus more on the functions of the twins in the virtual space which include Model Management, Service Management and Twin Management layers. They are integrated with the physical entities through

the Data Ingestion/Processing layer, and interact with users through the User Interaction layer on the top.

3.3.3.2. Compare with ISO 23247 Digital Twin Reference Architecture. The recent published standard 'ISO 23247, Automation systems and integration – Digital Twin framework for manufacturing' (ISO 2021) defines the principles and requirements for developing DTs in manufacturing domain, and provides a framework to support the creation of DTs of observable manufacturing elements (e.g. personnel, equipment, materials, facilities etc.). This standard represents the trend of standardisation for DT in different domains. Although mainly focused on the manufacturing domain, it provides valuable reference for DT standardisation in other domains as well as cross domains. It is worthy to compare the proposed CDT reference architecture with this standard.

The entity-based reference model of the ISO 23247 Digital Twin framework is shown in Figure 10. The lifecycle dimension is not considered in this standard, although in the Annex A of ISO 23247-3, it refers to ISO 10303-239 (ISO 2005) which specifies the application protocol for product life cycle support. As shown in Figure 10, both frameworks are composed of physical entities (Observable Manufacturing Elements) and digital entities in general. Different from the five-layer architecture in the proposed CDT framework, the ISO 23247 framework consists of four types of entities including User Entity, Core Entity, Data Collection and Device Control Entity,

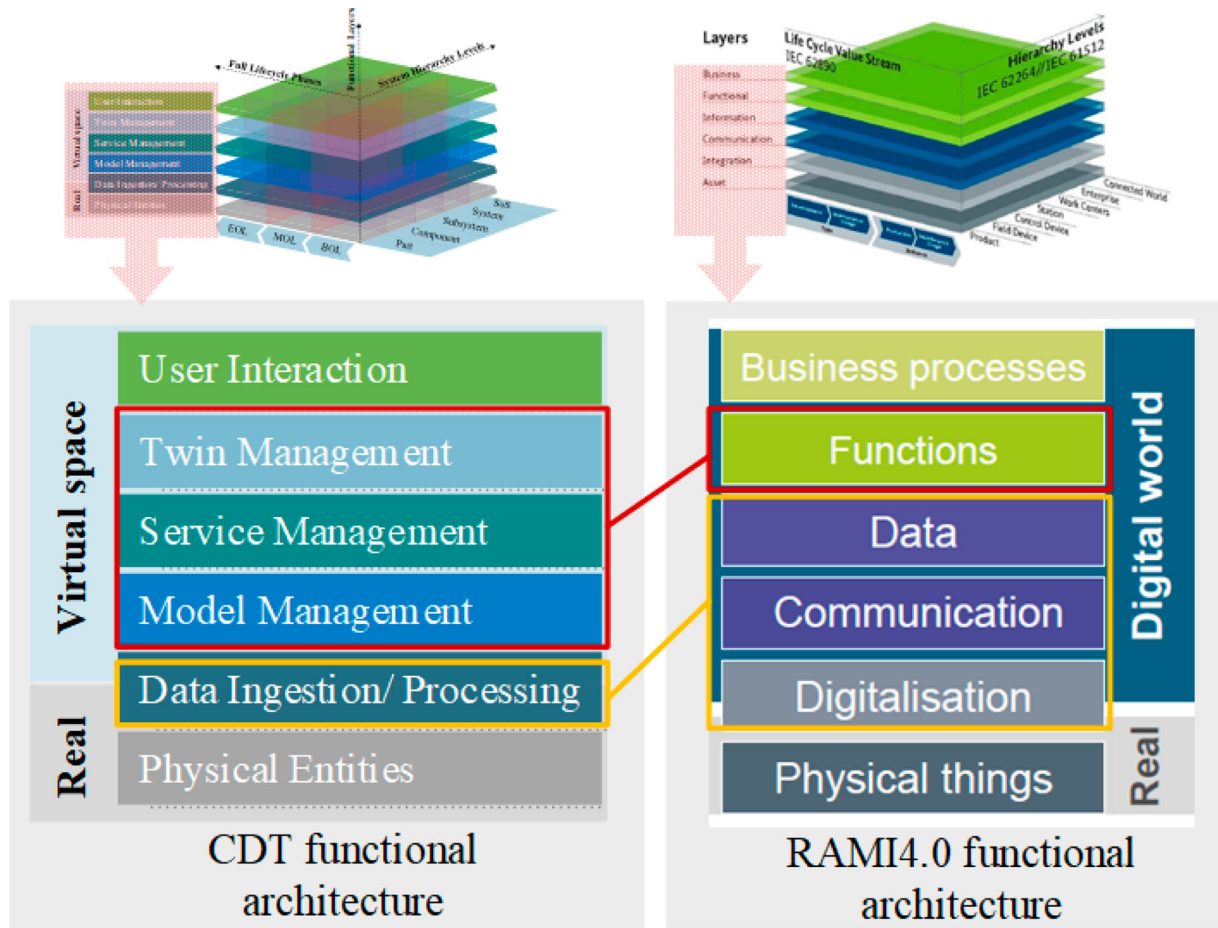


Figure 9. Comparison of functional architecture between CDT and RAMI4.0.

and Cross-System Entity. Among them, the Data Collection and Device Control Entity can be mapped to the Data Ingestion and Processing layer, whereas the other types of entities are different from the CDT framework functional layers. The most critical differences are the Twin Management and Model Management functions. Because the ISO 23247 framework is mainly aimed at a single DT model, the interaction and integration of multiple DTs is not included. In comparison, the CDT framework treats the interaction and integration of multiple DTs as the core function, which can be realised by the kg and Ontology module in the Model management layer, Service Orchestrator in the Service layer and the entire Twin Management layer.

3.4. Cognitive digital twin applications

As an emerging concept, CDT has not yet been widely implemented and verified in industry. Most published studies either explore the theoretical perspectives of CDT or focus on the CDT vision. However, there are several ongoing studies and projects which aim to verify the CDT feasibility by applying it to different industry scenarios.

The active EU project COGNITWIN (Cognitive plants through proactive self-learning hybrid digital twins) is a dedicated project aiming to enhance the cognitive capabilities of existing process control systems thus to enable self-organising and offering solutions to unpredicted behaviours. In a recent study (Abburu et al. 2020b), a cognitive twin toolbox conceptual architecture, as shown in Figure 11, was applied to several use cases from the process industry, including (1) operational optimisation of gas treatment centre (GTC) in aluminium production, (2) minimise health and safety risks and maximise the metallic yield in Silicon (Si) production, (3) condition monitoring of assets in steel and related products production, (4) real-time monitoring of finished products for operational efficiency, (5) improving heat exchanger efficiency in power production, chemicals, and food processing industries. As an example, CDT is used to empower the construction of a smart steel pipe production plant in one of the COGNITWIN project pilots. The detailed application background, objectives, key criteria and some primary results of this pilot has been introduced in Albayrak and Ünal (2020).

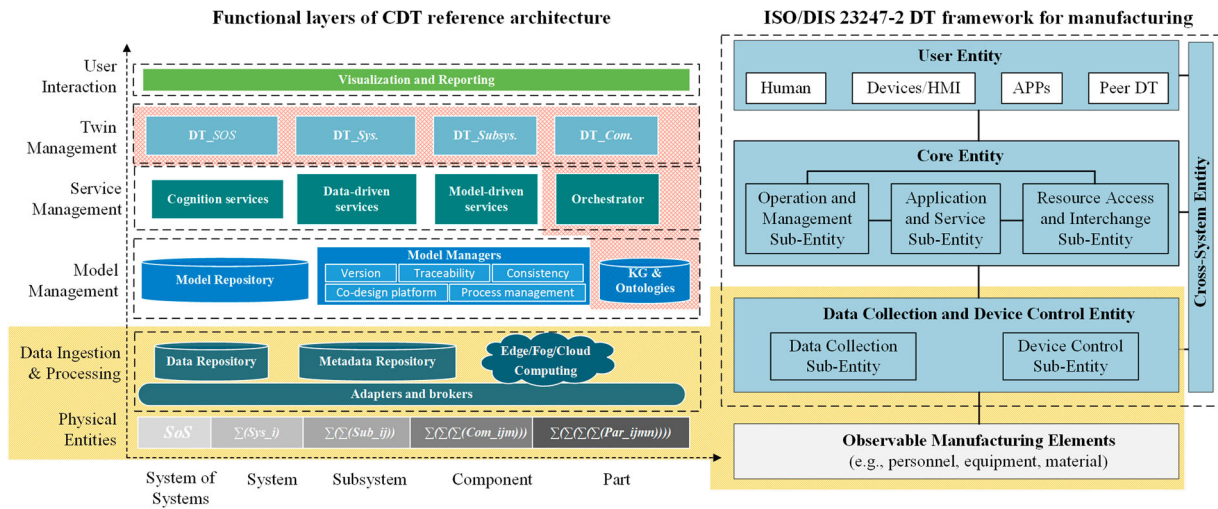


Figure 10. Comparison of CDT functional architecture and ISO/DIS 23247-2 DT framework for manufacturing.

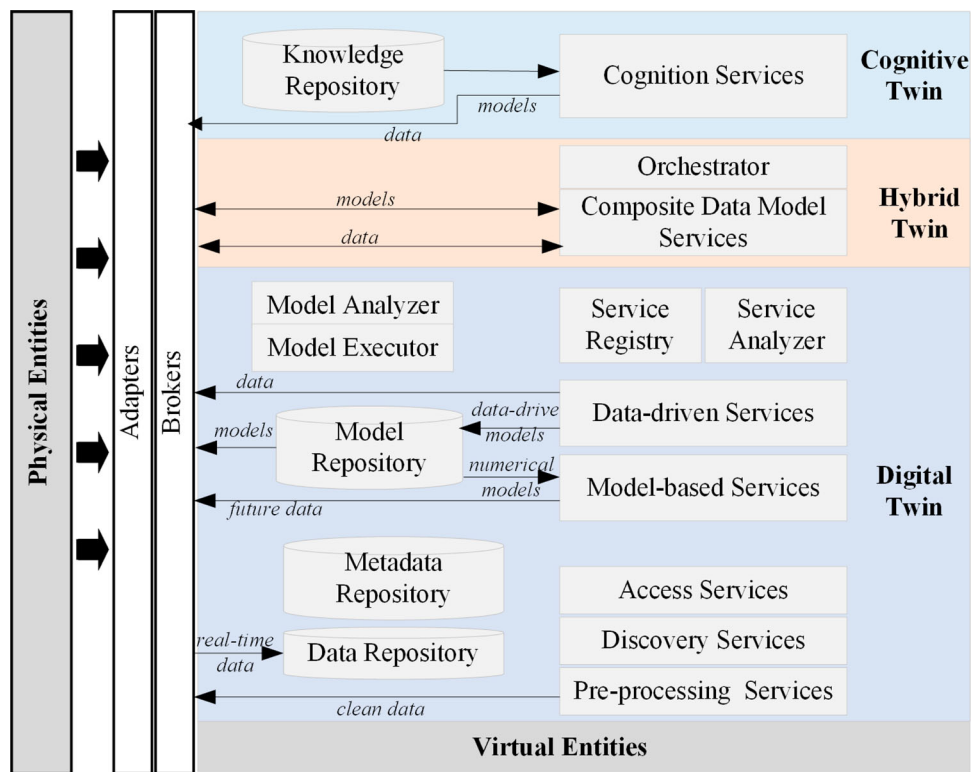


Figure 11. Cognitive Twin Toolbox conceptual architecture (adapted from Abburu et al. 2020b).

The ongoing EU project FACTLOG (Energy-aware Factory Analytics for Process Industries) aims to improve the cognition capabilities of complex process systems by combining data-driven and model-driven digital twins. CDT is one of the main enabling technologies in this project. Some primary results about CDT have been achieved. For example, Lu et al. (2021) used semantic modelling and systems engineering approaches to construct a cognitive approach for the complexity management of digital twin systems. Based on this work,

they developed a CDT application framework based on knowledge graph (Lu, Zheng et al. 2020). An application case has been presented in a recent study (Rožanec et al. 2021; Rožanec and Jinzhi 2020) as shown in Figure 12, in which the authors proposed a actionable cognitive twin enabled by knowledge graph to support demand forecasting and production planning in a manufacturing plant. This approach was implemented in a European original equipment manufacturer in automotive industry. The qualitative and quantitative analysis of

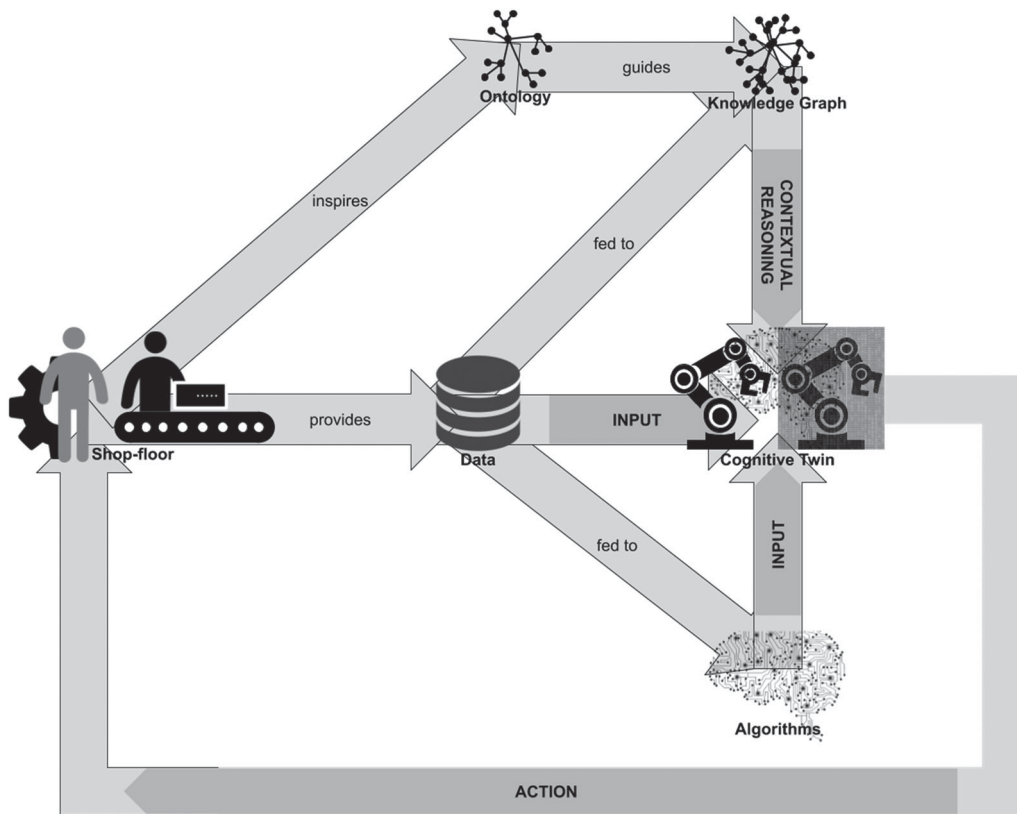


Figure 12. CDT for decision-makings in the manufacturing (Rozanec and Jinzhi 2020).

two use cases are used to verify the advantages of the proposed CDT approach.

In another EU project QU4LITY (Digital Reality in Zero Defect Manufacturing), which aims to provide open, standardised, and transformable zero-defect manufacturing product/service/model, CDT has been used as a semantically enhanced version of digital twin to support autonomous quality. To improve product quality, a data model named RMPFQ (Resource, Material, Process, Feature/Function, Quality) (Zheng, Psarommatidis et al. 2020) was adopted to support the ontology development. Based on these application scenarios, a CDT model is under construction for an aircraft assembly system in one of the QU4LITY project pilots. Some primary results have been achieved covering trade-off for assembling schedules, data flow across model-based systems engineering, semantic models, system architectures and co-simulation for verification (Zheng et al. 2021).

In addition to the ongoing projects, some existing studies also applied the CDTs in different industries. For example, Zhou et al. (2020) integrated dynamic knowledge bases with digital twin models to enable knowledge-based intelligent services for autonomous manufacturing. The knowledge-driven digital twin

supports intelligent perceiving, simulating, understanding, predicting, optimising and controlling strategy. This approach was applied to three use cases including manufacturing process planning, production scheduling and production process analysis. Moreover, Yitmen et al. (2021) make use of CDTs to support building information management. Through CDTs, process optimisation and decision-making are mainly supported based on knowledge graph modelling and reasoning across the entire building lifecycle.

Some commercial artificial intelligence platforms also provide industrial solutions to accelerate CDT applications. For instance, the IBM Watson IoT platform (Fariz Saracevic 2017) can provide cognitive analytic services to support analytics-for-design and design-for-analytics interactions. It is a close-loop framework including functions covering requirement management, system design, verify and validate, build and deploy, as well as operate etc. For example, it provides a Cognitive Requirements Advisor which enables semantic-aware requirement management to integrate and analyse requirements for design, manufacturing and operation phases. Its Cognitive Quality Advisor can help optimise manufacturing processes by analysing different performance indicators to maximise benefits and minimise defects and

costs etc. Based on the Watson IoT platform, they provides the cognitive manufacturing solutions (IBM 2021) composed of four functional layers, including device layer, IoT platform layer, application layer and industry solutions layer. It can serve as a implementation reference for CDT applications in process and quality improvement, resource optimisation and supply chain optimisation.

The analysis of CDT applications indicates that the main beneficiaries of CDT are complex production systems and processes that involve multiple subsystems with stakeholders from different domains or crossing multiple lifecycle phases. CDT can enable creating a unified framework to orchestrate the interactions among subsystems and processes. From the system requirement point of view, CDT provides solution for industrial systems that require higher level of agile, resilient and reconfigurable capabilities, as well as enhanced decision-making and autonomous reaction abilities.

4. Key enabling technologies

Advanced technologies, such as semantic technologies, IIoT, artificial intelligence etc., are the pillars for modern industrial systems. Since the CDT concept is built upon DT concept, all the enabling technologies of DT are essential for CDT as well. Some of them are particularly important for the CDT paradigm. This section introduces some of the key enabling technologies for CDT development.

4.1. Semantic technologies

Semantic technologies are considered as the core of the CDT concept due to their advantages for improving data interoperability and constructing cognitive capabilities. CDT models usually involve heterogeneous data, information and knowledge making it difficult to align among different DTs and stakeholders. Semantic technologies such as ontologies and knowledge graphs provide promising solutions for this challenge.

4.1.1. Ontology engineering

Ontology is considered as a promising core for cognitive systems owing to its capability of formalising the ontological features of the physical entities which is compatible with the perspective of human common sense (El Kadiri and Kiritsis 2015). Ontology engineering refers to a set of activities that concern the ontology development process and the ontology lifecycle, the methods and methodologies for building ontologies, and the tool suites

and languages that support them (Gruber 1993; El Kadiri and Kiritsis 2015).

Nowadays there exists various of mature methodologies, tools and languages to support ontology development. However, numerous ontologies have been created based on various languages and tools for different application scenarios. It becomes a challenging task to integrate different ontologies in a unified framework to assure their interoperability. Thus this is a common problem when developing CDT models for complex systems. A possible solution is to make use of a hierarchical methodology to unify the application ontologies under a common top-level ontology which contains a set of general vocabularies commonly used across all domains. These vocabularies are properly structured and formally defined according to certain methodology. Such top-level ontologies provide a common foundation for developing lower-level ontologies such as domain-specific ontologies and more detailed application ontologies. The adoption of the top-level ontology assures semantic interoperability among these lower-level ontologies. Currently, many top-level ontologies have been developed and widely applied by different communities such as Basic Formal Ontology (BFO) (Arp and Smith 2008) and Descriptive Ontology for Linguistic and Cognitive Engineering (DOLCE) (Masolo et al. 2003).

Some recent efforts have been spent on unifying and standardising existing domain ontologies based on certain top-level ontologies. For example, the Industrial Ontologies Foundry (IOF) (IOF 2021) which is an ongoing initiative aims to co-create a set of open ontologies to support the manufacturing for industrial needs and to promote data interoperability. IOF provides a multi-layer architecture to guide ontology development, consisting of four layers: top-level foundation ontology, domain-level (domain independent and domain specific) reference ontologies, subdomain ontologies and application ontologies. It uses BFO as a foundation, which experts from different industrial domains work jointly to create open and principles-based ontologies.

A top-level ontology enables the semantic interoperability among the lower-level ontologies in the same family. However, several top-level ontologies have been developed and each of them has their own followers. It is necessary to develop unified specifications for these top-level ontologies through cross-domain collaborations involving most relevant stakeholders. A set of standards and specifications are necessary to integrate existing and widely used top-level ontologies. Moreover, certain ontology alignment approaches are expected to maximise the use of existing domain ontologies developed under different top-level ontologies.

4.1.2. Knowledge graph

Knowledge graph is considered as the most promising enabler for realising CDT vision. It uses a graph model composed of nodes and edges to describe the topology of both structured and unstructured data enabling formal, semantic, and structured representation of knowledge (Nguyen, Vu, and Jung 2020). The nodes in a knowledge graph are used to represent entities or raw values encoded as literals; and connected edges are used to describe the semantic relations between nodes. Knowledge graph enables to construct semantic connections among heterogeneous data sources thus to capture underlying knowledge from them. This is especially important for the CDT vision where a large amount of data and models are involved.

Since Google released its knowledge graph for its searching engine in 2012 (Singhal 2012), knowledge graph and relevant technologies have attracted significant attentions from both academic researchers and industrial practitioners (Ehrlinger and Wöß 2016; Nguyen, Vu, and Jung 2020). Most of these studies are from information technology domain such as recommendation systems, question-answering systems, cybersecurity systems and semantic search systems etc. Nguyen, Vu, and Jung (2020). The applications of knowledge graph in production sector are relatively less. In some recent studies (Banerjee et al. 2017; Boschert, Heinrich, and Rosen 2018; Gómez-Berbís and de Amescua-Seco 2019; Lu, Zheng et al. 2020), it has been used to enhance digital twins which is one of the main driving forces for constructing CDT concept as introduced in the related work. In many cases, knowledge graph is applied together with ontologies to create a knowledge base or knowledge management system. A typical approach is to use knowledge graph to acquire information from raw data and to integrate the information into an ontology where a reasoner can be executed to derive new knowledge (Ehrlinger and Wöß 2016; Nguyen, Vu, and Jung 2020).

Within the CDT reference architecture, knowledge graph and ontologies are the backbone for the functional layers of data ingestion and processing, model management, service management and twin management. They are the core to integrate different models from different systems, subsystems and components across different lifecycle phases.

4.2. Model-based systems engineering

Systems Engineering provides a set of principles and concepts, as well as scientific, technological, and management methods to support realisation, use and retirement of engineered systems (Sillitto et al. 2019). It is an efficient approach to manage highly complex

industrial systems which is the target of CDTs. More specifically, Model-based systems engineering (MBSE) enables the formalism of system architectures to support tasks such as model-based requirement elicitation, specification, development and testing etc. Lu, Ma et al. (2020). It emerged around 1993 not only from industry (Oliver 1993) but also from academia (Wymore 2018), which is proposed for supporting complex system development such as aerospace. MBSE is a key enabling technology corresponding to the formalism of system hierarchy levels of the CDT reference architecture. It can be used to formalise the hierarchy of a complex industrial system and specify the relationships among digital models from different hierarchy levels.

Currently, MBSE is widely used to support digital twin development and integration, and cognition construction for digital twins. For example, Liu et al. (2021) developed a shop floor digital twin based on MBSE. Bachelor et al. (2020) proposed an MBSE approach for digital twin development and integration in the aerospace industry using SysML and linked data. Moreover, Lu, Zheng et al. (2020) proposed a CDT structure based on a systems engineering methodology and demonstrated how MBSE can facilitate the cognition construction during CDT development.

4.3. Product lifecycle management

As presented in the reference architecture, a CDT should support the integration of digital models across different phases of entire lifecycle. Product Lifecycle Management (PLM) plays a critical role on the lifecycle management. PLM is a strategic approach for managing product related information efficiently during its whole product lifecycle, including BOL, MOL and EOL (Kiritis 2011). Its main objectives include universal and secure management of product information, maintaining the integrity of that product information throughout the product lifecycle and management of business processes for creating, managing, disseminating and using the information.

Nowadays, PLM has been widely used in different industrial sectors, especially in manufacturing domain, as an important strategy to maintain the sustainable and competitive advantages of enterprises (Liu and Liang 2015; Zhang et al. 2017). PLM has also been impacted by many emerging technologies such as IoT, data mining and semantic technologies etc. These technologies makes possible of capturing hidden knowledge and patterns from massive lifecycle data, thus to improve various data-driven services (Denkena, Schmidt, and Krüger 2014), which aligns with the CDT vision. These applications and advancements provide substantial basis for CDT development.

4.4. Industrial data management technologies

IIoT is a subset of IoT with applications in industrial domain by deploying large number of smart things in industrial systems to enable real-time sensing, collecting, and processing. IIoT systems usually require higher levels of security and reliable communications to assure high production performances (Liao, Loures, and Deschamps 2018; Khan et al. 2020). The rapid development of IIoT and relevant technologies has been one of the main driving forces for Industry 4.0 and smart manufacturing, which is also the foundation of CDT. The big data generated by IIoT devices provides input for the data ingestion and processing layer of the CDT reference architecture for constructing data-driven services as basis.

During the past decade, large amount of efforts from both academia and industry have been spent on IIoT technologies and their applications thanks to the sufficient investment all over the globe. The achievements of these efforts provide a starting point for CDT development which is a matter of properly utilising and reusing these relevant technologies when adopting them to the CDT vision. The wide deployment of IIoT devices in modern industrial systems generates large volume of industrial data with different data formats and structure. These big industrial data is the basis for all data-driven services of CDT. On the other hand, collection, storage, sharing and processing of these data remain a challenging task which requires multiple advanced technologies.

4.4.1. Cloud/Fog/Edge computing

Cloud computing has been used as one of the most important tools to cope with the big data challenge in many sectors. It has become an essential digital infrastructure for many industrial enterprises where a mature market has been created. Companies such as Amazon, Google, IBM, and Microsoft etc., are providing flexible computing services and solutions using their own cloud computing platforms (Qi and Tao 2019). However, the communication between the cloud server and local data sources requires high bandwidth leading to heavy delay which is not acceptable in many application cases especially for CDT vision where real-time processing is critical.

To tackle this problem, fog computing and edge computing have been proposed to bring the cloud computing closer to data generators and consumers (Mohan and Kangasharju 2016). Fog and edge computing are considered as extensions of cloud computing (Roman, Lopez, and Mambo 2018). Fog computing allows network devices run cloud applications on their native architecture (Solutions, Cisco Fog Computing 2015), while

edge computing uses edge devices such as tablets, smart-phones, nano data centres and single board computers etc. as a cloud to perform basic computing tasks (Chandra, Weissman, and Heintz 2013; Ordieres-Meré, Villalba-Díez, and Zheng 2019).

Fog and edge computing enables transferring part of the computing, storage, and networking capabilities of cloud to the local network which makes possible of low latency, real time response, reduction of network traffic, etc. Qi and Tao (2019) and Wang et al. (2021). These characteristics make them essential components for intelligent industrial systems. They are key enabling tools for the physical entity layer and the data ingestion and processing layer of the proposed CDT reference architecture when developing the basic bridges between physical world and digital world.

4.4.2. Natural language processing

Natural language processing (NLP) is a complementary tool for semantics technologies and machine learning during the CDT development. NLP aims to gather knowledge following a similar way that human beings understand and use language based on a set of appropriate tools and techniques (Chowdhury 2003). In a CDT system, both structured and unstructured data are involved. Among them, a large part of the raw data are stored with natural languages such as technical documents, logs, orders and etc. It is required to transform these information into computer-understandable data so that they can be integrated with other data sources which makes it possible to create a comprehensive digital model. Currently, NLP has been a popular research topic for several decades as a subfield of linguistics, computer science, and artificial intelligence. There are numerous NLP algorithms and techniques available for performing tasks like text and speech processing, optical character recognition, morphological analysis and syntactic analysis etc. Similar to machine learning, it is also a matter of selecting suitable solutions based on the specific requirements of the CDT application scenarios.

4.4.3. Distributed ledger technology

A CDT system requires integration of data from different stakeholders. The main concerns during data sharing are about data security and privacy and intellectual property (IP) protection. To deal with these concerns, reliable cybersecurity infrastructure and data encryption mechanisms are necessary. Boschert, Heinrich, and Rosen (2018) proposed two ways to control the degree of transparency of CDTs, i.e. using encapsulated models to guarantee IP protection and open models for realising integrated development processes. The recent advancement of Distributed Ledger Technologies (DLT),

including blockchain, provides a decentralised solution for protecting data security and privacy during data sharing. DLT removes the dominant administrator and central database compared with traditional data sharing approaches which enables secure data sharing in a trust-less environment. It has attracted increasing attention from both researchers and practitioners recently. Various DLT systems and platforms have been developed including private ledgers, permissioned ledgers and public ledgers etc. As an example, Sun et al. (2020) developed a IIoT data handling architecture using a public ledger and IOTA Tangle (Popov 2018; Chen et al. 2020), to ensure data privacy and to protect data ownership in a distributed platform. Considering the advantages of DLT, it will be a promising technology to accelerate the CDT vision and should be taken into consideration during future CDT development.

5. Challenges and opportunities

As an emerging concept, CDT is still in an very early stage of its development. There are many challenges to be resolved in order to fully realise its vision. Although the aforementioned enabling technologies are supposed to cope with some of the challenges, much more efforts are needed in the future. This section summarises a few main challenges for CDT which also could be potential opportunities for the future studies.

5.1. Knowledge management

The most challenging task for CDT is to realise the cognitive capability. A functional and comprehensive knowledge base is the core for this capability. In a recent paper, Abburu et al. (2020a) summarised the challenges for CDT cognition into three sections, i.e. knowledge representation, knowledge acquisition and knowledge update.

- (1) **Knowledge representation:** it refers to how to specify and to formalise the relevant information and knowledge so that digital models can be used to enable intelligent activities as input. The main difficulties are the heterogeneity and ambiguity of such knowledge. Semantic technologies and machine learning can address this challenge. As Abburu et al. (2020a) suggested, ontologies can be used for constructing the domain knowledge and rules for representing the problem-solving knowledge. Machine learning algorithms can analyse big industrial data and formalise the hidden information and patterns into knowledge. It is also important to take into account the standardisation of the knowledge to assure interoperability.

- (2) **Knowledge acquisition:** it refers to how to collect and process human understandable knowledge into a machine understandable format. It is difficult because a large part of knowledge is stored in documents, logs and reports etc., which are written in natural languages. Sometimes it is even stored in multimedia formats such as images, audios and videos etc. It is a complex and time-consuming process to process such knowledge sources and to acquire knowledge from them. To deal with this challenge, NLP technology could provide possible solutions such as text mining, speech recognition and OCR techniques etc. Data mining also provides solutions to discover hidden knowledge from raw data following the data-information-knowledge flow.
- (3) **Knowledge update:** it refers to how to continuously update the existing knowledge and derive new knowledge. It is difficult because it is a complex process consisting of knowledge extension, knowledge forgetting and knowledge evolution. It is challenging not only because of the difficulties to ensure the consistency after applying a change, but also more importantly to discover the need for a change at the right time. Abburu et al. (2020a) suggests two possible solutions: usage-driven strategies based on experts' feedback; and structure-driven methods based on ontology reasoners to detect conflicting rules.

5.2. Integration of DT models

Many DT models have been created separately by different stakeholders of a complex system corresponding to its different subsystems or components across the entire lifecycle. These existing DT models need to be integrated and orchestrated properly into the CDT architecture enabled by the twin management layer. It is a challenging task in real applications due to the fact that different stakeholders might adopted different standards, protocols and structures for their DT models. This leads to the challenges of interoperability issues at data level. Moreover, this brings the DT models with more complex features and services for integration.

Semantic technologies and systems engineering provide possible solutions for addressing this challenge. Advanced semantic tools, such as knowledge graph, enable to semantically link digital models as individual entities and to use edges to represent their interrelationships. Systems engineering and MBSE, provide a series of methodologies and tools to manage complex system models systematically. For example, Wang, Wang et al. (2019) proposed a MBSE method to structure a

complex system into hierarchical layers, i.e. meta-meta models, meta-models and models, which are then represented using a unified approach named GOPPRR (Graph, Object, Point, Property, Role, and Relationship). Through, this approach, DT models can be described by unified ontology models.

5.3. Standardisation

Standardisation is fundamental to achieve interoperability of data and tool for constructing CDT. The CDT development requires to support integration of data and digital twin models at different functional layers. Therefore, the formal standardisation of DTs is the basis for enabling the CDT paradigm. Several Standards Developing Organisations (SDOs) are developing DT standards such as the International Organization for Standardization (ISO), World Wide Web Consortium – Web of Things (W3C WoT), the Industrial Internet Consortium (IIC) and the Plattform Industrie 4.0, etc. Jacoby and Usländer (2020). It will be a challenging task to unify and align the relevant DT standards developed by different SDOs in the future. Some efforts have been spent to discuss the alignment between these existing and upcoming standards but no formalised and structured agreement has been published yet (Jacoby and Usländer 2020).

Despite the lack of a universal DT standard, some existing standards and protocols can be considered as substitute. For example, Plattform Industrie 4.0 provides the Asset Administration Shell (AAS) as a part of the RAMI4.0 (Schweichhart 2016). ETSI Industry Specification Group (ISG) proposes the Next Generation Service Interfaces-Linked Data (NGSI-LD) APIs (Alliance, Open Mobile 2012). The W3C WoT working group proposes the WoT ThingDescription (WoT TD) which is an official W3C Recommendation (W3C-WoT 2020). Such standards and APIs are discussed in previous studies (Jacoby and Usländer 2020; Abburu et al. 2020b). The recent published ISO 23247 standard defines the principles and requirements for developing DTs in manufacturing domain, and provides a framework to support the creation of DTs of observable manufacturing elements including personnel, equipment, materials, manufacturing processes, facilities, environment, products, and supporting documents (ISO 2021).

5.4. Implementations

Compared with DTs, the implementation of CDTs is much more challenging. DTs usually focus only on one physical system during one of its lifecycle phases. Therefore, DTs can be implemented inside the enterprises

supported by certain technical providers when necessary. However, a CDT might include multiple physical systems covering several lifecycle phases with stakeholders from different enterprises. The implementation of CDTs thus requires both intra- and inter-organisation collaborations. In addition to the above mentioned interoperability issue, it brings new challenges in terms of project management, data privacy/security concerns, and IP protection etc. The lack of successful demonstrators of CDT implementation further increases its risks.

To mitigate the risks during CDT implementation, the following strategies could be considered. From managerial perspective, the CDT development and implementation should be treated as a complex project organised by experienced experts and following correct project management methodologies. Lessons learned from similar complex projects, such as PLM implementation (Batenburg, Helms, and Versendaal 2006; Hewett 2010), should be taken into account. For example, it is recommended to follow a step-wise approach, starting from a mature DT system, then increasing the number of digital models and covering more lifecycle phases.

From technical perspective, the above mentioned systems engineering and DLT might also provide some inspirations. Systems engineering enables to guide the the implementation using a system thinking methodology to capture the system nature by analysing the interrelationships between the components within the system boundary (Haskins 2014). DLT enables to create a decentralised platform for eliminating concerns about data privacy/security and IP protection (Zheng, Lu et al. 2020) in order to encourage inter-organisational collaborations.

The implementation of CDT requires a series of enabling technologies. The couplings of these technologies is another challenging task of CDT applications. From functional perspective, it is crucial to design a reliable CDT architecture to orchestrate the interactions of different functional components. For this purpose, some systems engineering methodologies, such as MBSE, and related approaches like GOPPRR can be used to support the design of system architectures. From cyber-physical system perspective, it is critical to interconnect subsystems across different domains and lifecycle phases using adapters, brokers and other types of middleware mechanisms. The aforementioned standards and specifications such as AAS and NGSI-LD provide references for creating such coupling mechanisms.

6. Limitations and future work

As an emerging concept, CDT represents a trend of an important evolution of DT. CDT is still in its initial

stage of concept development and there are many challenges and open issues to be addressed to fully realise its vision. In this position paper, we firstly reviewed existing studies relevant to the CDT concept. Based on the review results, we summarised the main characteristics of CDT and proposed a new definition. To explain the CDT vision and facilitate its development, we designed a three-dimensional CDT reference architecture based on an RAMI4.0 architecture from previous studies. This reference architecture contains the key elements of the CDT vision including full lifecycle phases, system hierarchical level and multi-layer functions. Moreover, we introduced the main enabling technologies for CDT development and implementation directly correlated with the CDT characteristics, including the semantic technologies, MBSE and PLM etc. To pave ways for future studies, we analysed the challenges of CDT development and application. Considering the importance and application of DTs in the past years, we believe CDT, as the next evolution of DT, will attract more and more attention from both academia and industry. Moreover, CDT is expected to become another powerful tool for realising the intelligent manufacturing paradigm.

As an exploratory study, there are several limitations in this paper. Firstly, as a novel concept, there is still lack of a widely agreed definition of CDT. Academic and industrial experts have proposed CDT definitions from various perspectives. This study tried to extract the commons elements from existing definitions. However, due to the limited efforts and resources, we are not able to include all existing studies. Secondly, the implementation of CDT requires to integrate a series of advanced technologies. Currently, there are few successful implementation cases that fully realised the vision of CDT. The definition and reference architecture proposed in this study need to be verified through the aforementioned ongoing projects and more future works. Moreover, the technical details about the enabling technologies and their implementations in the CDT paradigm are not discussed in this paper. More research efforts are needed to explore these technologies and bridge the gaps between conceptual framework and industrial applications.

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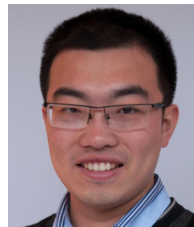
Data availability statement

The authors confirm that the data supporting the findings of this study are available within the article.

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References

- Abburu, Sailesh, A. J. Berre, M. Jacoby, D. Roman, L. Stojanovic, and Nenad Stojanovic. 2020a. "Cognitive Digital Twins for the Process Industry." In *COGNITIVE 2020, The Twelfth International Conference on Advanced Cognitive Technologies and Applications*, 68–73. Think Mind.
- Abburu, Sailesh, Arne J. Berre, Michael Jacoby, Dumitru Roman, Ljiljana Stojanovic, and Nenad Stojanovic. 2020b. "COGNITWIN-Hybrid and Cognitive Digital Twins for the Process Industry." In *2020 IEEE International Conference on Engineering, Technology and Innovation (ICE/ITMC)*, 1–8. IEEE.
- Adl, Ahmed El. 2016. *The Cognitive Digital Twins: Vision, Architecture Framework and Categories*. Technical Report. https://www.slideshare.net/slideshow/embed_code/key/JB60Xqcn.

- Albayrak, Özlem, and Perin Ünal. 2020. "Smart Steel Pipe Production Plant via Cognitive Digital Twins: A Case Study on Digitalization of Spiral Welded Pipe Machinery." In *Cybersecurity workshop by European Steel Technology Platform*, 132–143. Springer.
- Al Faruque, Mohammad Abdullah, Deepan Muthirayan, Shih-Yuan Yu, and Pramod P Khargonekar. 2021. "Cognitive Digital Twin for Manufacturing Systems." In *2021 Design, Automation & Test in Europe Conference & Exhibition (DATE)*, 440–445. IEEE.
- Ali, Muhammad Intizar, Pankesh Patel, John G. Breslin, Ramy Harik, and Amit Sheth. 2021. "Cognitive Digital Twins for Smart Manufacturing." *IEEE Intelligent Systems*, 36 (2): 96–100. doi:10.1109/MIS.2021.3062437.
- Alliance, Open Mobile. 2012. *Next Generation Service Interfaces Architecture*. Technical Report. http://www.openmobilealliance.org/release/NGSI_S/V1_0-20120.
- Arp, Robert, and Barry Smith. 2008. "Function, Role, and Disposition in Basic Formal Ontology." *Nature Precedings*, 1–1. <https://doi.org/10.1038/npre.2008.1941.1>.
- Bachelor, Gray, Eugenio Brusa, Davide Ferretto, and Andreas Mitschke. 2020. "Model-Based Design of Complex Aeronautical Systems Through Digital Twin and Thread Concepts." *IEEE Systems Journal* 14 (2): 1568–1579. <https://ieeexplore.ieee.org/document/8835120/>
- Banerjee, Agniva, Raka Dalal, Sudip Mittal, and Karuna Pande Joshi. 2017. "Generating Digital Twin models using Knowledge Graphs for Industrial Production Lines." In *Workshop on Industrial Knowledge Graphs, Co-Located With the 9th International ACM Web Science Conference 2017*.
- Batenburg, R., R. W. Helms, and J. Versendaal. 2006. "PLM Roadmap: Stepwise PLM Implementation Based on the Concepts of Maturity and Alignment." *International Journal of Product Lifecycle Management* 1 (4): 333–351.
- Boschert, Stefan, Christoph Heinrich, and Roland Rosen. 2018. "Next Generation Digital Twin." In *Proc. tmce*, 209–218. Las Palmas de Gran Canaria, Spain.
- Chandra, Abhishek, Jon Weissman, and Benjamin Heintz. 2013. "Decentralized Edge Clouds." *IEEE Internet Computing* 17 (5): 70–73.
- Chen, Jingjing, Tiefeng Cai, Wenxiu He, Lei Chen, Gang Zhao, Weiwen Zou, and Lingling Guo. 2020. "A Blockchain-Driven Supply Chain Finance Application for Auto Retail Industry." *Entropy* 22 (1): 95. <https://www.mdpi.com/1099-4300/22/1/95>
- Cho, Sangje, Gökan May, and Dimitris Kiritsis. 2019. "A Semantic-Driven Approach for Industry 4.0." In *2019 15th International Conference on Distributed Computing in Sensor Systems (DCOSS)*, 347–354. IEEE.
- Chowdhury, Gobinda G. 2003. "Natural Language Processing." *Annual Review of Information Science and Technology* 37 (1): 51–89.
- Denkena, Berend, Justin Schmidt, and Max Krüger. 2014. "Data Mining Approach for Knowledge-based Process Planning." *Procedia Technology* 15: 406–415.
- Ding, Kai, Felix T. S. Chan, Xudong Zhang, Guanghui Zhou, and Fuqiang Zhang. 2019. "Defining a Digital Twin-based Cyber-physical Production System for Autonomous Manufacturing in Smart Shop Floors." *International Journal of Production Research* 57 (20): 6315–6334.
- Duprez, Jean. 2019. "Approach to Structure, Formalize and Map MBSE Meta-models and Semantic Rules." *INCOSE International Symposium* 29 (1): 22–36. <https://onlinelibrary.wiley.com/doi/abs/10.1002/j.2334-5837.2>
- Ehrlinger, Lisa, and Wolfram Wöß. 2016. "Towards a Definition of Knowledge Graphs." *SEMANTiCS (Posters, Demos, SuCESS)* 48: 1–4.
- Eirinakis, Pavlos, Kostas Kalaboukas, Stavros Lounis, Ioannis Mourtos, Jože M. Rožanec, Nenad Stojanovic, and Georgios Zois. 2020. "Enhancing Cognition for Digital Twins." In *2020 IEEE International Conference on Engineering, Technology and Innovation (ICE/ITMC)*, 1–7. IEEE.
- El Kadiri, Soumaya, and Dimitris Kiritsis. 2015. "Ontologies in the Context of Product Lifecycle Management: State of the Art Literature Review." *International Journal of Production Research* 53 (18): 5657–5668.
- Fariz Saracevic, I. B. M. 2017. *Cognitive Digital Twin*. Technical Report. <https://www.slideshare.net/BosniaAgile/cognitive-digital-twin>.
- Fernández, Felipe, Ángel Sánchez, José F. Vélez, and A. Belén Moreno. 2019. "Symbiotic Autonomous Systems with Consciousness Using Digital Twins." In *International Work-Conference on the Interplay Between Natural and Artificial Computation*, 23–32. Springer.
- Gockel, Brian, Andrew Tudor, Mark Brandyberry, Ravi Penmettsa, and Eric Tuegel. 2012. "Challenges With Structural Life Forecasting Using Realistic Mission Profiles." In *53rd AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference 20th AIAA/ASME/AHS Adaptive Structures Conference 14th AIAA*, 1813.
- Gómez-Berbis, Juan Miguel, and Antonio de Amescua-Seco. 2019. "SEDIT: Semantic Digital Twin Based on Industrial IoT Data Management and Knowledge Graphs." 178–188. http://link.springer.com/10.1007/978-3-030-34989-9_14.
- Grieves, Michael. 2014. "Digital Twin: Manufacturing Excellence Through Virtual Factory Replication." *White Paper* 1: 1–7.
- Gruber, Thomas R. 1993. "A Translation Approach to Portable Ontology Specifications." *Knowledge Acquisition* 5 (2): 199–220.
- Haskins, Cecilia. 2014. "A Journey Through The Systems Landscape." *Insight (American Society of Ophthalmic Registered Nurses)* 17 (2): 63–64. <http://doi.wiley.com/10.1002/inst.201417263a>
- Hewett, Andrew. 2010. "Product Lifecycle Management (PLM): Critical Issues and Challenges in Implementation." In *Information Technology and Product Development*, 81–105. Springer.
- IBM. 2021. *Cognitive Manufacturing: An Overview and Four Applications that are Transforming Manufacturing Today*. Technical Report. <https://www.ibm.com/downloads/cas/VDNKMWM6>.
- IOF. 2021. *IOF Technical Principles*. Technical Report. <https://www.industrialontologies.org/technical-principles/>.
- ISO. 2005. *ISO 10303-239:2005 Industrial Automation Systems and Integration – Product Data Representation and Exchange – Part 239: Application Protocol: Product Life Cycle Support*. Technical Report. <https://www.iso.org/standard/38310.html>.
- ISO. 2021. *ISO 23247-1:2021 Automation Systems and Integration – Digital Twin Framework for Manufacturing – Part 1: Overview and General Principles*. Technical Report. <https://www.iso.org/standard/75066.html>.

- Jacoby, Michael, and Thomas Usländer. 2020. "Digital Twin and Internet of Things—Current Standards Landscape." *Applied Sciences* 10 (18): 6519.
- Khan, Wazir Zada, M. H. Rehman, Hussein Mohammed Zangoti, Muhammad Khalil Afzal, Nasrullah Armi, and Khaled Salah. 2020. "Industrial Internet of Things: Recent Advances, Enabling Technologies and Open Challenges." *Computers & Electrical Engineering* 81: 106522.
- Kharlamov, Evgeny, Francisco Martin-Recuerda, Brandon Perry, David Cameron, Roar Fjellheim, and Arild Waaler. 2018, December. "Towards Semantically Enhanced Digital Twins." In *2018 IEEE International Conference on Big Data (Big Data)*, 4189–4193. IEEE. <https://ieeexplore.ieee.org/document/8622503/>.
- Kiritsis, Dimitris. 2011. "Closed-loop PLM for Intelligent Products in the Era of the Internet of Things." *Computer-Aided Design* 43 (5): 479–501.
- Knapp, G. L., Tuhin Mukherjee, J. S. Zuback, H. L. Wei, T. A. Palmer, Amitava De, and T. J. A. M. DebRoy. 2017. "Building Blocks for a Digital Twin of Additive Manufacturing." *Acta Materialia* 135: 390–399.
- Kokar, Mieczyslaw M., and Mica R. Endsley. 2012. "Situation Awareness and Cognitive Modeling." *IEEE Intelligent Systems* 27 (3): 91–96.
- Kossiakoff, Alexander, William N. Sweet, Samuel J. Seymour, and Steven M. Biemer. 2011. *Systems Engineering: Principles and Practices*. Vol. 83. John Wiley & Sons.
- Liao, Yongxin, Eduardo de Freitas Rocha Loures, and Fernando Deschamps. 2018. "Industrial Internet of Things: A Systematic Literature Review and Insights." *IEEE Internet of Things Journal* 5 (6): 4515–4525.
- Liu, Yang, and Liting Liang. 2015. "Evaluating and Developing Resource-based Operations Strategy for Competitive Advantage: An Exploratory Study of Finnish High-tech Manufacturing Industries." *International Journal of Production Research* 53 (4): 1019–1037.
- Liu, Juan, Jianhua Liu, Cunbo Zhuang, Ziwen Liu, and Tian Miao. 2021. "Construction Method of Shop-floor Digital Twin Based on MBSE." *Journal of Manufacturing Systems* 60 (April): 93–118. <https://doi.org/10.1016/j.jmsy.2021.05.004>
- Lu, Jinzhi, Junda Ma, Xiaochen Zheng, Guoxin Wang, and Dimitris Kiritsis. 2020. "Design Ontology Supporting Model-based Systems-engineering Formalisms." *arXiv preprint arXiv:2010.07627*.
- Lu, Jinzhi, Xiaochen Zheng, Ali Gharaei, Kostas Kalaboukas, and Dimitris Kiritsis. 2020. "Cognitive Twins for Supporting Decision-Makings of Internet of Things systems." In *Proceedings of 5th International Conference on the Industry 4.0 Model for Advanced Manufacturing*, 105–115. Springer.
- Lu, Jinzhi, Xiaochen Zheng, Lukas Schweiger, and Dimitris Kiritsis. 2021. "A Cognitive Approach to Manage the Complexity of Digital Twin Systems." In *Smart Services Summit*, 105–115. Springer.
- Masolo, C., S. Borgo, A. Gangemi, N. Guarino, A. Oltramari, and L. Schneider. 2003. "Dolce: A Descriptive Ontology for Linguistic and Cognitive Engineering." *WonderWeb Project, Deliverable D17 v2* 1: 75–105.
- Matthews, Sky. 2018. *Designing Better Machines: The Evolution of a Cognitive Digital Twin explained*. Technical Report. <https://www.ibm.com/blogs/internet-of-things/iot-evolution-of->
- Mohan, Nitinder, and Jussi Kangasharju. 2016. "Edge-Fog cloud: A distributed Cloud for Internet of Things Computations." In *2016 Cloudification of the Internet of Things (CIoT)*, 1–6. IEEE.
- Muller, Gerrit. 2008. "A Reference Architecture Primer." *Eindhoven Univ. of Techn., Eindhoven, White paper*.
- Nguyen, Hoang Long, Dang Thinh Vu, and Jason J. Jung. 2020. "Knowledge Graph Fusion for Smart Systems: A Survey." *Information Fusion* 61: 56–70.
- Nickel, Maximilian, Kevin Murphy, Volker Tresp, and Evgeniy Gabrilovich. 2015. "A Review of Relational Machine Learning for Knowledge Graphs." *Proceedings of the IEEE* 104 (1): 11–33.
- Oliver, David W. 1993. "Descriptions of Systems Engineering Methodologies and Comparison of Information Representations." *INCOSE International Symposium* 3 (1): 97–104. <https://onlinelibrary.wiley.com/doi/abs/10.1002/j.2334-5837.1>
- Ordieres-Meré, Joaquín, Javier Villalba-Díez, and Xiaochen Zheng. 2019. "Challenges and Opportunities for Publishing IIoT Data in Manufacturing As a Service Business." *Procedia Manufacturing* 39: 185–193.
- Popov, S. 2018. *The Tangle, IOTA Whitepaper*. Technical Report. IOTA, Tech. Rep. [Online]. Available: <https://iota.org/IOTAWhitepaper.pdf....>
- Psarommatis, Foivos. 2021. "A Generic Methodology and a Digital Twin for Zero Defect Manufacturing (ZDM) Performance Mapping Towards Design for ZDM." *Journal of Manufacturing Systems* 59: 507–521. <https://www.sciencedirect.com/science/article/pii/S0278612521>
- Qi, Qinglin, and Fei Tao. 2019. "A Smart Manufacturing Service System Based on Edge Computing, Fog Computing, and Cloud Computing." *IEEE Access* 7: 86769–86777.
- Qi, Qinglin, Fei Tao, Tianliang Hu, Nabil Anwer, Ang Liu, Yongli Wei, Lihui Wang, and A. Y. C. Nee. 2019. "Enabling Technologies and Tools for Digital Twin." *Journal of Manufacturing Systems* 58: 3–21. <https://linkinghub.elsevier.com/retrieve/pii/S027861251930086>.
- Roman, Rodrigo, Javier Lopez, and Masahiro Mambo. 2018. "Mobile Edge Computing, Fog et al.: A Survey and Analysis of Security Threats and Challenges." *Future Generation Computer Systems* 78: 680–698.
- Rožanec, Joze M., and Lu Jinzhi. 2020. "Towards Actionable Cognitive Digital Twins for Manufacturing." In *International Workshop on Semantic Digital Twins co-located with ESWC, SeDiT@ESWC*.
- Rožanec, Jože M., Jinzhi Lu, Jan Rupnik, Maja Škrjanc, Dunja Mladenčić, Blaž Fortuna, Xiaochen Zheng, and Dimitris Kiritsis. 2021. "Actionable Cognitive Twins for Decision Making in Manufacturing." *International Journal of Production Research*, 1–27. <https://doi.org/10.1080/00207543.2021.2002967>
- Schleich, Benjamin, Nabil Anwer, Luc Mathieu, and Sandro Wartzack. 2017. "Shaping the Digital Twin for Design and Production Engineering." *CIRP Annals* 66 (1): 141–144.
- Schweichhart, Karsten. 2016. "Reference Architectural Model Industrie 4.0 (Rami 4.0)." *An Introduction*. Available online: <https://www.plattform-i40.de/I>

- Seshadri, Banavara R., and Thiagarajan Krishnamurthy. 2017. "Structural Health Management of Damaged Aircraft Structures Using Digital Twin Concept." In *25th AIAA/AHS Adaptive Structures Conference*, 1675.
- Sillitto, Hillary, James Martin, Dorothy McKinney, Regina Griego, Dov Dori, Daniel Krob, Patrick Godfrey, Eileen Arnold, and Scott Jackson. 2019. "Systems Engineering and System Definitions." In *INCOSE*.
- Singhal, Amit. 2012. *Introducing the Knowledge Graph: Things, not Strings*. Technical Report. <https://blog.google/products/search/introducing-knowledge-gra>.
- Söderberg, Rikard, Kristina Wärmefjord, Johan S. Carlson, and Lars Lindkvist. 2017. "Toward a Digital Twin for Real-time Geometry Assurance in Individualized Production." *CIRP Annals* 66 (1): 137–140.
- Solutions, Cisco Fog Computing. 2015. "Unleash the Power of the Internet of Things."
- Sun, Shengjing, Xiaochen Zheng, Javier Villalba-Díez, and Joaquín Ordieres-Meré. 2020. "Data Handling in Industry 4.0: Interoperability Based on Distributed Ledger Technology." *Sensors* 20 (11): 3046.
- Tao, Fei, Fangyuan Sui, Ang Liu, Qinglin Qi, Meng Zhang, Boyang Song, Zirong Guo, Stephen C.-Y. Lu, and Andrew Y. C. Nee. 2019. "Digital Twin-driven Product Design Framework." *International Journal of Production Research* 57 (12): 3935–3953.
- Tao, Fei, Meng Zhang, Jiangfeng Cheng, and Qinglin Qi. 2017. "Digital Twin Workshop: a New Paradigm for Future Workshop." *Computer Integrated Manufacturing Systems* 23 (1): 1–9.
- Tao, Fei, He Zhang, Ang Liu, and A. Y. C. Nee. 2019. "Digital Twin in Industry: State-of-the-Art." *IEEE Transactions on Industrial Informatics* 15 (4): 2405–2415. <https://ieeexplore.ieee.org/document/8477101/>
- W3C-WoT. 2020. *Web of Things (WoT) Thing Description - W3C Recommendation 9 April 2020*. Technical Report. <https://www.w3.org/TR/wot-thing-description/>.
- Wang, J., Z. Chen, S. Guo, Y. Li, and Z. Lu. 2021. "Optimal Sprinting Pattern in Thermal Constrained CMPs." *IEEE Transactions on Emerging Topics in Computing* 9 (1): 484–495.
- Wang, Xi Vincent, and Lihui Wang. 2019. "Digital Twin-based WEEE Recycling, Recovery and Remanufacturing in the Background of Industry 4.0." *International Journal of Production Research* 57 (12): 3892–3902.
- Wang, Hongwei, Guoxin Wang, Jinzhi Lu, and Changfeng Ma. 2019. "Ontology Supporting Model-Based Systems Engineering Based on a GOPRR Approach." In *World Conference on Information Systems and Technologies*, 426–436. Springer.
- Wang, Jinjiang, Lunkuan Ye, Robert X. Gao, Chen Li, and Laibin Zhang. 2019. "Digital Twin for Rotating Machinery Fault Diagnosis in Smart Manufacturing." *International Journal of Production Research* 57 (12): 3920–3934.
- Wymore, A. Wayne. 2018. *Model-based Systems Engineering*, Vol. 3. CRC press.
- Yitmen, Ibrahim, Sepehr Alizadehsalehi, İlknur Akıner, and Muhammed Ernur Akıner. 2021. "An Adapted Model of Cognitive Digital Twins for Building Lifecycle Management." *Applied Sciences* 11 (9): 4276. <https://www.mdpi.com/2076-3417/11/9/4276>.
- Zhang, Yingfeng, Shan Ren, Yang Liu, Tomohiko Sakao, and Donald Huisinigh. 2017. "A Framework for Big Data Driven Product Lifecycle Management." *Journal of Cleaner Production* 159: 229–240.
- Zheng, Xiaochen, Jinzhi Lu, Rebeca Arista, Xiaodu Hu, Joachim Lentes, Fernando Ubis, Jyri Sorvari, and Dimitris Kiritsis. 2021. "Development of an Application Ontology for Knowledge Management to Support Aircraft Assembly System Design." In *FOMI 2021: 11th International Workshop on Formal Ontologies meet Industry, held at JOWO 2021: Episode VII The Bolzano Summer of Knowledge, September 11–18, 2021, Bolzano, Italy*, 1–9.
- Zheng, Xiaochen, Jinzhi Lu, Shengjing Sun, and Dimitris Kiritsis. 2020. "Decentralized Industrial IoT Data Management Based on Blockchain and IPFS." In *IFIP International Conference on Advances in Production Management Systems*, 222–229. Springer.
- Zheng, Xiaochen, Foivos Psarommatis, Pierluigi Petrali, Claudio Turrin, Jinzhi Lu, and Dimitris Kiritsis. 2020. "A Quality-Oriented Digital Twin Modelling Method for Manufacturing Processes Based on A Multi-Agent Architecture." *Procedia Manufacturing* 51: 309–315.
- Zhou, Guanghui, Chao Zhang, Zhi Li, Kai Ding, and Chuang Wang. 2020. "Knowledge-driven Digital Twin Manufacturing Cell Towards Intelligent Manufacturing." *International Journal of Production Research* 58 (4): 1034–1051.